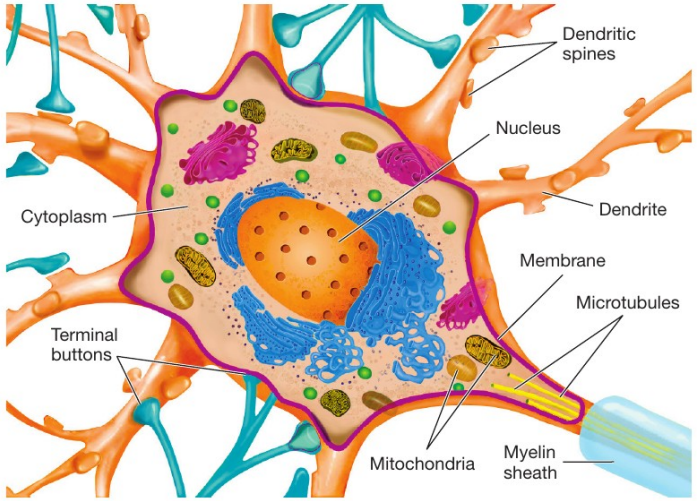
Figure 2.8 Internal Structures of a Neuron



Carlson and Birkett (2022), Fig. 2.8, p. 42.

BEHAVIORAL NEUROSCIENCE 392

Foundations and Frontiers

How Nerve Cells Work: From Electrical Signals to Chemical Communication

Fall 2026

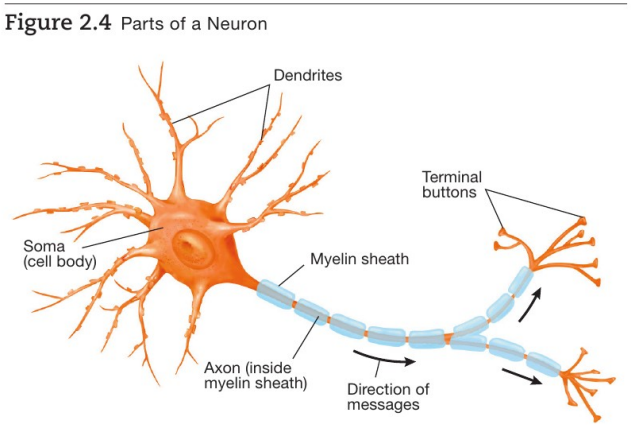
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

Neurons divide communication among specialized regions

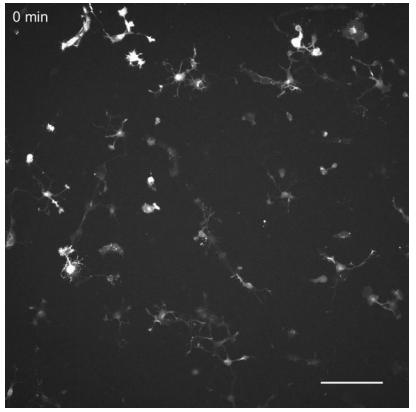
A **neuron** is a cell specialized for receiving and transmitting information. Its **soma**, or cell body, contains the nucleus and much of the machinery that maintains the cell. Branching **dendrites** receive many incoming signals, while an **axon** carries electrical signals toward distant targets.

The axon ends in **terminal buttons**, which release chemical messengers onto another cell. A receiving cell changes its activity when those messengers bind to receptors. This arrangement allows a signal to travel through a sequence of cells without requiring the cells to share one continuous membrane.

Neurons have different shapes because they receive inputs and contact targets in different places. A long axon can carry information far from its cell body. Many short dendritic branches can receive inputs locally. The location of each compartment affects where a signal enters and where it leaves.



Carlson and Birkett (2022), Fig. 2.4, p. 40.



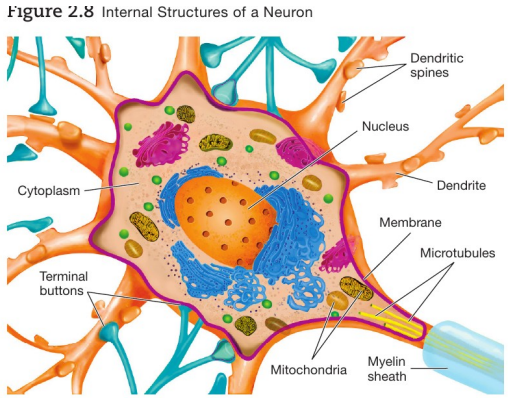
Winans et al. (2016), Video 1. Aesthetic context: Neurite growth and transport.

The cell body maintains distant neuronal compartments

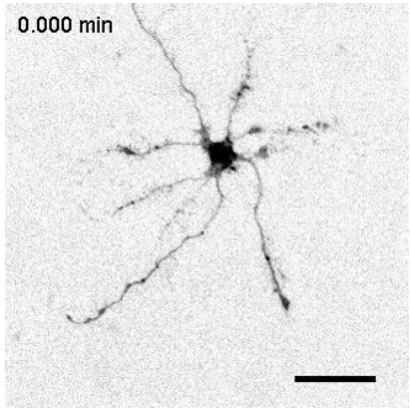
The **nucleus** contains DNA, the cell's genetic material. Cells use information in DNA to make proteins that form receptors, channels, enzymes, and structural components. The **endoplasmic reticulum** and **Golgi apparatus** help manufacture, modify, and package proteins that the neuron needs in its different regions.

An axon can extend far beyond the cell body, but its terminals still need membrane proteins and other materials. These materials must travel along the axon. The soma therefore maintains distant structures through an internal transport system rather than making every component separately at each terminal.

Mitochondria supply energy that supports cellular work, including maintaining ion gradients and moving cargo. A living neuron continually renews its components. Electrical signaling depends on this slower maintenance because channels, membranes, and the machinery that releases neurotransmitter must remain available where the cell uses them.



Carlson and Birkett (2022), Fig. 2.8, p. 42.



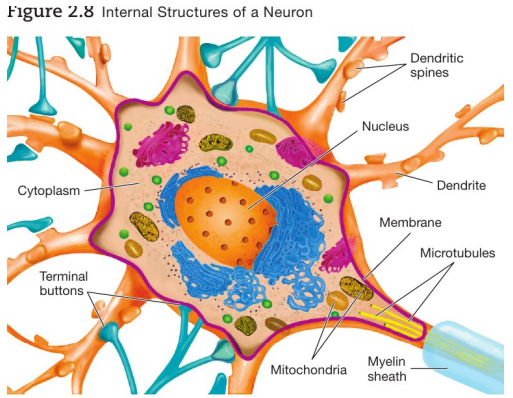
Winans et al. (2016), Video 3. Aesthetic context: Neurite growth and transport.

The cytoskeleton supports branches and carries cargo

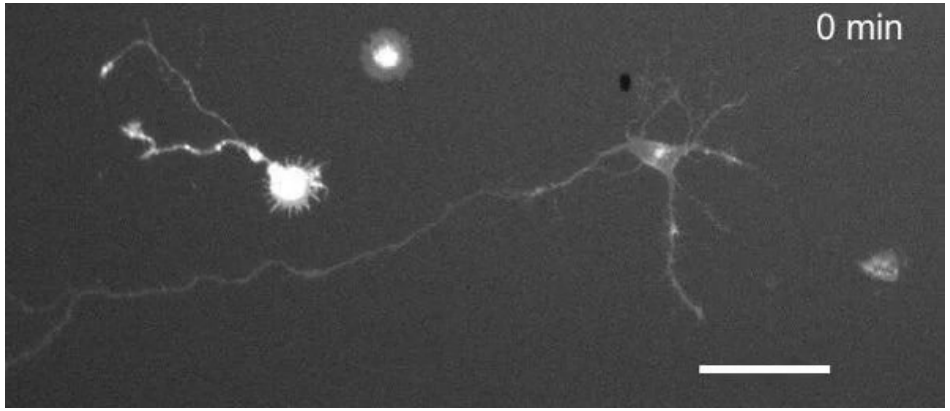
The **cytoskeleton** is an internal framework of protein structures. **Microtubules** help maintain the shape of neuronal processes and provide tracks for transport. **Actin** forms another part of this framework and participates in changes in cell shape, including movements at growing neuronal processes.

Axoplasmic transport moves materials through an axon. Motor proteins carry cargo along microtubules, using cellular energy to move. Transport toward the terminal is called **anterograde** transport. Transport toward the cell body is **retrograde** transport. Both directions help maintain communication between distant cellular compartments.

Moving a vesicle along an axon is different from conducting an action potential. The vesicle carries physical material through the cell. The action potential is an electrical event produced by changing ion movement across successive regions of membrane. Material transport maintains the structures that make electrical signaling possible.



Carlson and Birkett (2022), Fig. 2.8, p. 42.



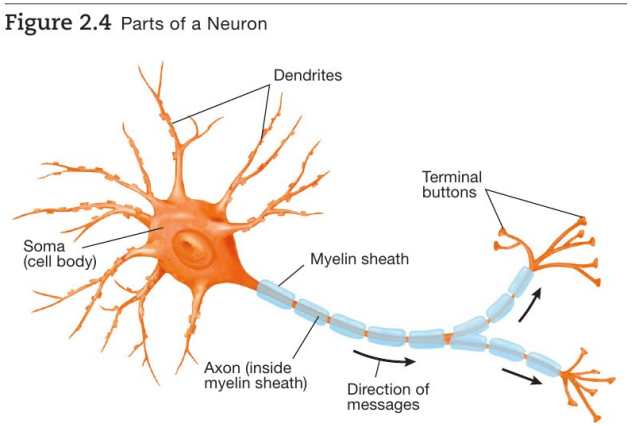
Winans et al. (2016), Video 9. Aesthetic context: Neurite growth and transport.

Dendrites provide surfaces for incoming synaptic signals

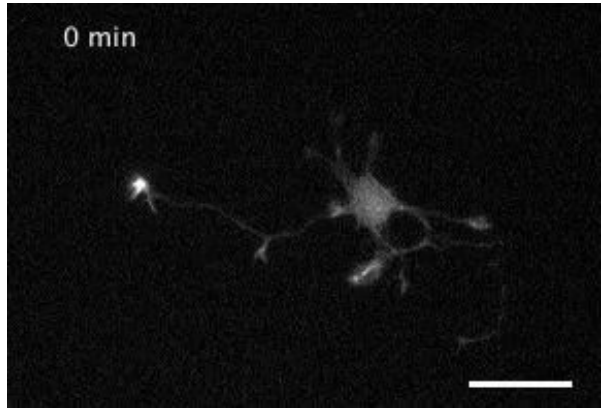
Dendrites extend from the soma and receive inputs from other neurons. Many dendrites contain small protrusions called **dendritic spines**. A spine can receive a synapse from an axon terminal. Branches and spines distribute receiving sites across a larger surface than the soma alone could provide.

Signals entering a dendrite change the voltage across its membrane. These local voltage changes can spread toward other parts of the neuron. Their influence depends on their size, location, and timing. Several incoming signals can therefore affect whether the neuron produces an outgoing action potential.

A developing neuron's branches must grow before they can occupy their mature positions. Microscopic recordings of growing processes provide cellular context for these structures. The movement of a growing branch occurs on a much slower timescale than the rapid electrical signals that a mature neuron conducts.



Carlson and Birkett (2022), Fig. 2.4, p. 40.



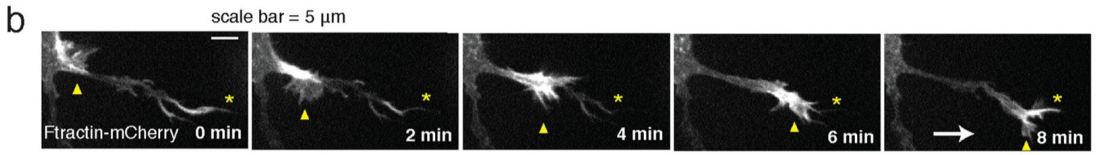
Winans et al. (2016), Video 2. Aesthetic context: Neurite growth and transport.

Actin waves accompany bursts of neurite growth

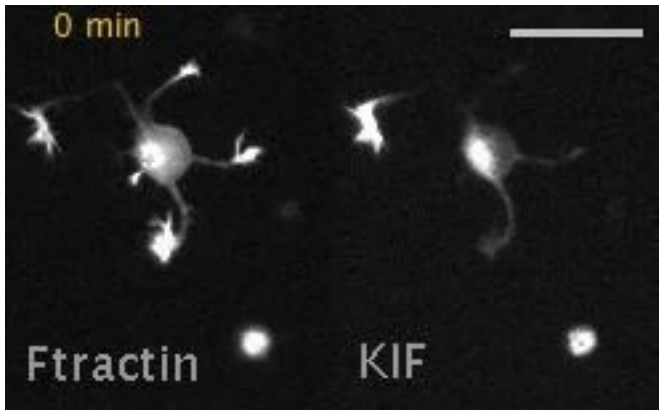
Winans and colleagues studied developing rat hippocampal neurons in culture. An immature projection is called a **neurite** before its mature identity is established. Fluorescent labels allowed the researchers to follow actin-rich waves as they moved outward from the cell body through these developing processes.

The actin waves traveled toward neurite tips, and neurite extension occurred in bursts associated with their arrival. Growing neurons did not extend every branch at one constant rate. Their changing internal organization produced periods of stronger growth in particular processes while other processes grew less.

Actin is therefore more than a permanent support beam. Its organization changes as a neuron develops. The authors proposed that outward-moving waves help a neurite explore its surroundings by changing growth and transport locally. This developmental process contributes to how the future receiving and sending compartments take shape.



Winans et al. (2016), Fig. 1b. Developing neurites, actin, and intracellular transport.



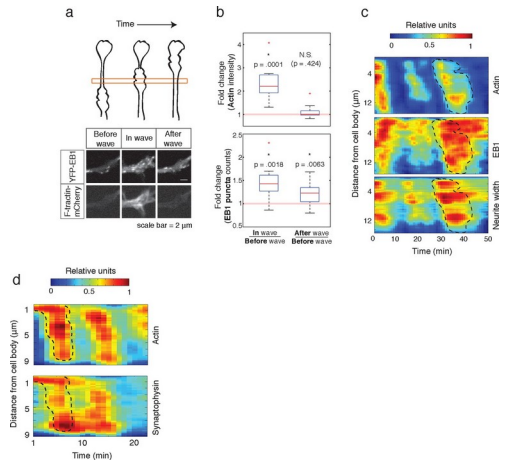
Winans et al. (2016), Video 5. Academic: Neurite growth and transport.

Microtubule growth increases within actin waves

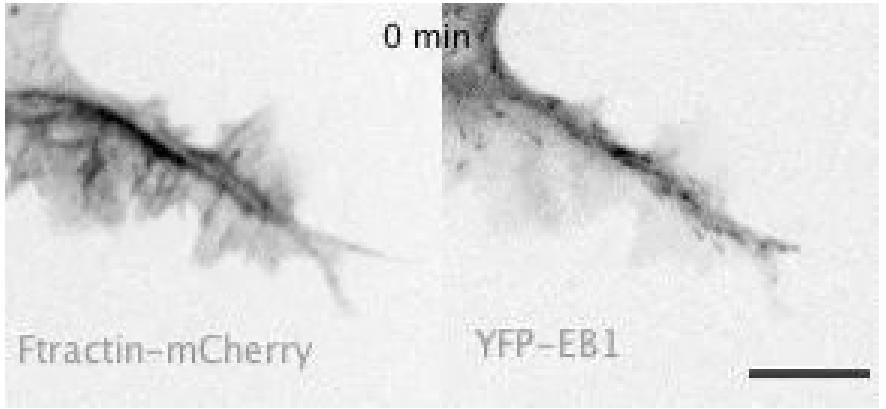
The researchers labeled growing microtubule ends with a fluorescent protein marker. They also labeled actin so both structures could be followed in the same developing neuron. These measurements tested whether an advancing actin wave coincided with changes in the microtubules used for transport.

Actin waves coincided with wider neurite regions and increased numbers of growing microtubule ends. The internal transport tracks changed as the wave passed. The authors proposed that transient widening creates room for additional microtubule growth, which increases the capacity for motor-driven transport through that region.

The interaction also depended on microtubule polymerization, the process of adding material to a microtubule. Actin and microtubules therefore influenced one another during neurite development. Their coordinated changes offer a cellular explanation for why growth occurs in pulses rather than as uninterrupted extension of every branch.



Winans et al. (2016), Fig. 9. Developing neurites, actin, and intracellular transport.



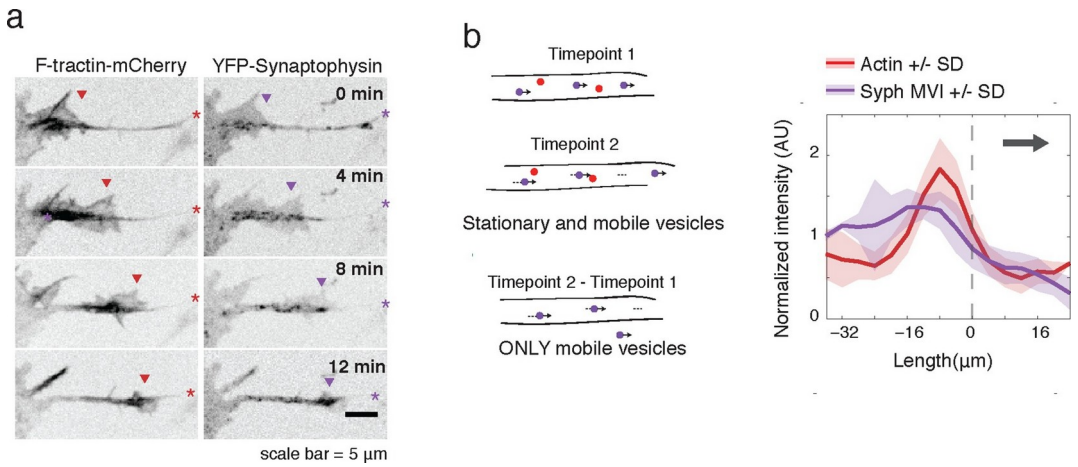
Winans et al. (2016), Video 7. Academic: Neurite growth and transport.

Motor-driven vesicle movement differs from diffusion

Winans and colleagues labeled vesicles containing **synaptophysin**, a protein associated with synaptic vesicles. They followed these fluorescent puncta within developing neurites. This approach allowed them to compare movement of membrane cargo with the position of an actin wave traveling through the same cellular process.

Vesicle movement within the wave differed from the random motion expected for particles freely diffusing in solution. Transport depended on the organized machinery inside the neurite. A vesicle can be directed along a track rather than simply wandering until it happens to reach a distant location.

An axon terminal needs proteins and membrane components that originate elsewhere in the neuron. Directed transport supplies those materials to the growing process. The results place membrane cargo, molecular motors, and the cytoskeleton in one physical pathway that prepares a neuron for communication with its target cells.



Winans et al. (2016), Fig. 4. Developing neurites, actin, and intracellular transport.

F-tractin-mCherry (static)

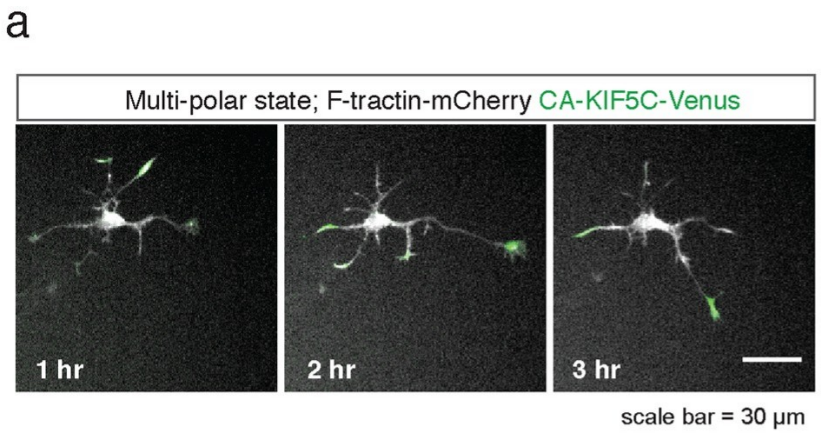
Winans et al. (2016), Video 6. Academic: Neurite growth and transport.

Developing neurons redistribute transport among branches

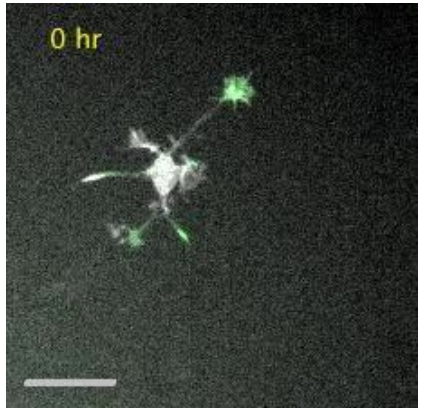
The study followed a fluorescently labeled kinesin motor construct as developing neurons produced several neurites. **Kinesin** is a family of proteins that moves along microtubules. The labeled construct allowed researchers to monitor changes in the distribution of motor-associated transport before one process became the dominant axon.

Transport signals shifted between neurites during the early developmental stage. Later, the signal became concentrated in one process. The neuron's internal traffic therefore changed as its initially similar projections acquired different identities. A mature axon emerged from a cell whose early branches had competed for growth and cargo.

The authors proposed that actin-driven fluctuations help redistribute transport during this competition. The proposal concerns how neuronal shape develops, rather than how an already mature neuron chooses an action. Building a polarized cell establishes the physical compartments that will later receive inputs and transmit signals in different directions.



Winans et al. (2016), Fig. 3a. Developing neurites, actin, and intracellular transport.



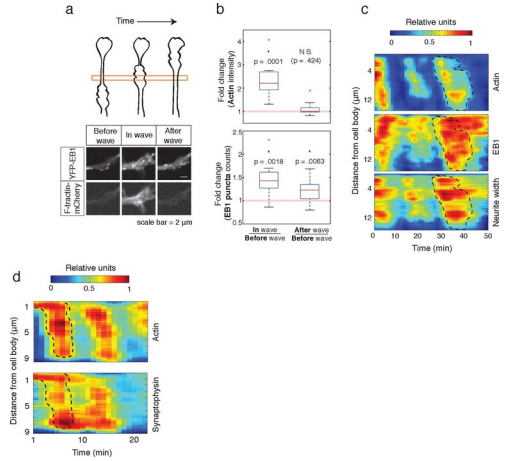
Winans et al. (2016), Video 4. Academic: Neurite growth and transport.

Growing processes accumulate membrane cargo locally

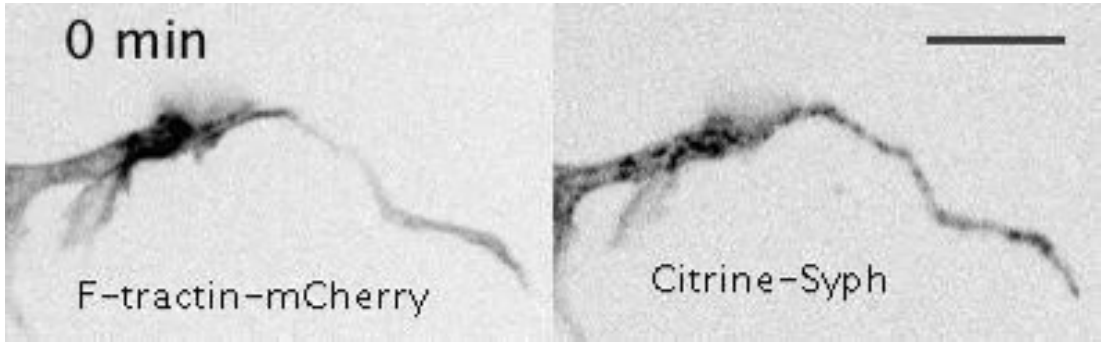
Fluorescent synaptophysin marked membrane cargo moving through immature neuronal processes. The researchers imaged this cargo together with actin waves. They could therefore determine whether a region undergoing a structural change also contained different amounts of material used to build and maintain neuronal membranes.

Actin waves coincided with transient increases in synaptophysin-positive vesicles. The growing process did not receive cargo uniformly over time. Regions with an advancing wave temporarily accumulated more material, placing vesicle traffic beside the cytoskeletal changes that accompanied increased microtubule growth and neurite widening.

A growing neurite needs both structural support and membrane material. Delivering cargo where growth is occurring can help supply the expanding region. The study describes coordinated changes in transport and shape, while the authors' mechanical model explains how those changes could promote the extension of one process.



Winans et al. (2016), Fig. 9. Developing neurites, actin, and intracellular transport.



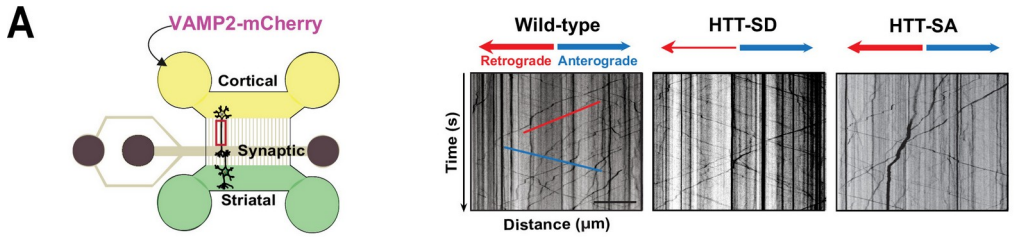
Winans et al. (2016), Video 8. Academic: Neurite growth and transport.

Axonal cargo delivery can change synaptic performance

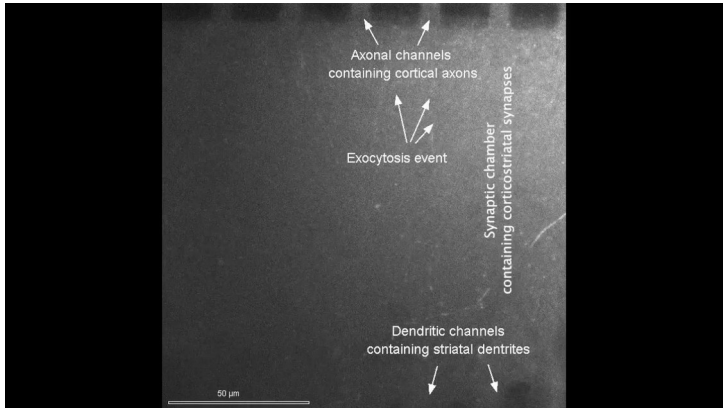
A later study examined how the protein **huntingtin** regulates transport of synaptic vesicle precursors in mouse neurons. These precursors carry components needed at release sites. Researchers studied the pathway in compartmentalized cultures and in the corticostriatal circuit, where cortical neurons send axons to the striatum.

Phosphorylation, the addition of a phosphate group to a protein, altered huntingtin's recruitment of the motor **KIF1A**. Increased recruitment accelerated precursor transport and increased synaptic **glutamate** release. Glutamate is a neurotransmitter that often excites receiving neurons. Changing a transport-regulating protein therefore changed the delivery of materials and the performance of a distant synapse.

Mice with persistently altered huntingtin phosphorylation accumulated excess vesicles and learned a rotating-rod motor task poorly. Reducing KIF1A restored transport and learning toward normal levels. The experiment connects a specific intracellular transport mechanism to synaptic release and the acquisition of a practiced motor skill.



Vitet et al. (2023), Fig. 1A. Axonal transport and synaptic function.



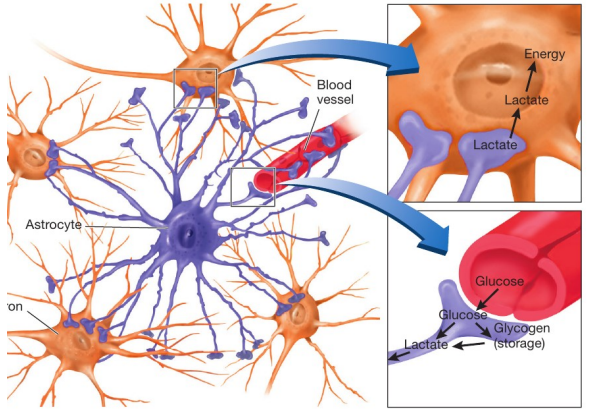
Vitet et al. (2023), Figure 1, video 1. Academic: Synaptic glutamate release.

Glia perform distinct jobs around neurons

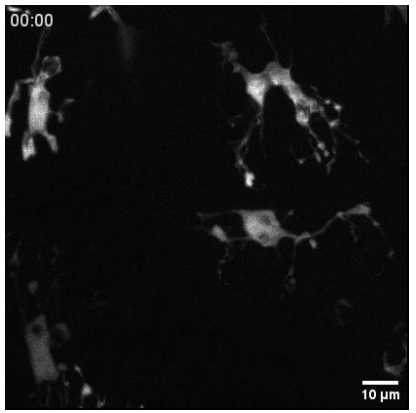
Glia are non-neuronal cells that maintain and regulate the nervous system. They are not one uniform population. **Astrocytes** participate in the chemical environment around neurons, **oligodendrocytes** form central myelin, and **microglia** act as resident immune cells within nervous tissue.

These cell types affect different parts of neuronal communication. An axon needs an appropriate extracellular environment, an intact membrane, and adequate metabolic resources. Glial activity helps maintain these conditions. A neuron can therefore have the correct internal channels yet function poorly when surrounding cells fail to perform their normal work.

Living brain tissue also changes over time as cells grow, die, and remove debris. Microscopy of glial cells provides context for this active maintenance. Microglial movement is not a neuronal action potential; it is a cellular behavior that helps preserve the environment in which neuronal signals occur.



Carlson and Birkett (2022), Fig. 2.10, p. 44.



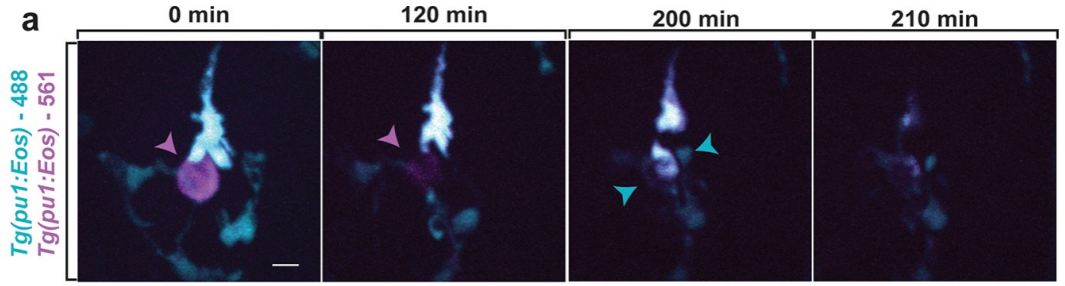
Gordon et al. (2024), pbio.3002819.s006.avi. Aesthetic context: Microglial cell dynamics.

Microglia engulf material from damaged or dying cells

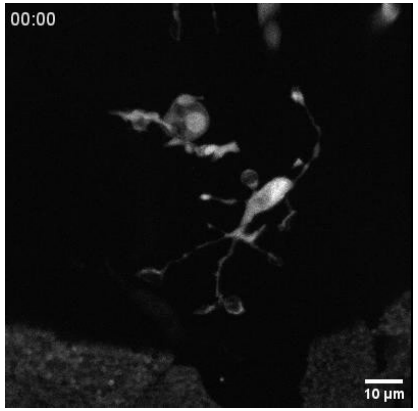
Microglia participate in the nervous system's response to cellular damage and debris. Through **phagocytosis**, a cell surrounds and takes in material from outside itself. Material inside the cell can then be processed by its degradative machinery. This activity differs from sending a neurotransmitter signal across a synapse.

A microglial cell can alter its shape and extend or retract processes while interacting with nearby tissue. Its behavior affects the surroundings of neurons and other glia. Maintaining the nervous system therefore involves both rapid electrical communication and slower cellular actions that remove unwanted material from the local environment.

Real time-lapse recordings make the changing positions of these cells visible. The intervals between captured frames can compress hours into seconds of playback. The source recording's timing must be retained and identified, because the apparent speed on screen is different from the rate of the biological process.



Gordon et al. (2024), Fig. 3a. Microglial turnover and debris handling.



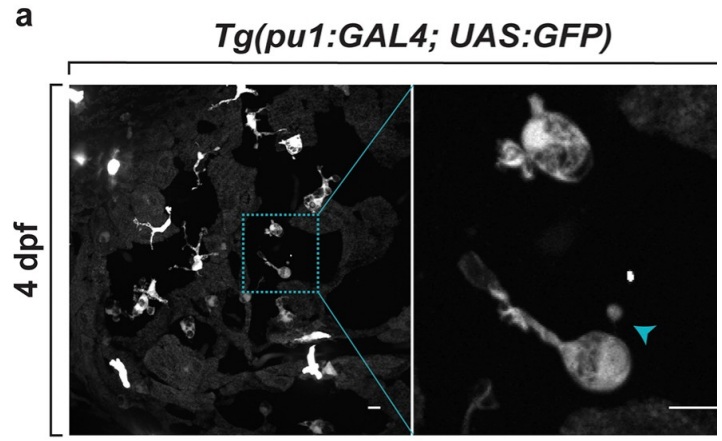
Gordon et al. (2024), pbio.3002819.s007.avi. Aesthetic context: Microglial cell dynamics.

Cell fragments must be distinguished from intact cells

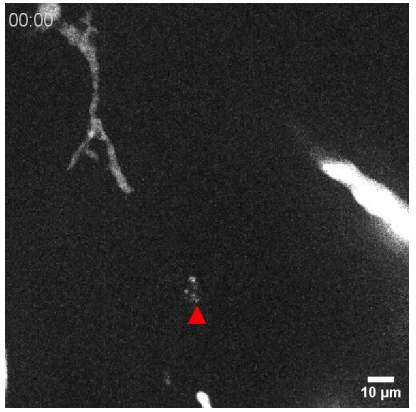
A 2024 study tracked microglia in the developing zebrafish brain. Researchers used fluorescent genetic labels to identify microglial material during brain colonization. A labeled spot could represent an intact cell or debris, so they examined cellular form and followed the same regions over time.

The investigators observed microglial cells dying while the overall population was expanding. Fluorescent fragments remained after a cell lost its intact form. Population growth therefore did not mean that every original microglial cell survived. New cells and cell loss occurred during the same developmental period.

Following cells over time changed the interpretation of a fluorescent label. A stationary fragment and a migrating intact cell could carry the same marker while representing different biological states. The experiment identifies cell turnover as part of brain development rather than treating every labeled object as a functioning immune cell.



Gordon et al. (2024), Fig. 1a. Microglial turnover and debris handling.



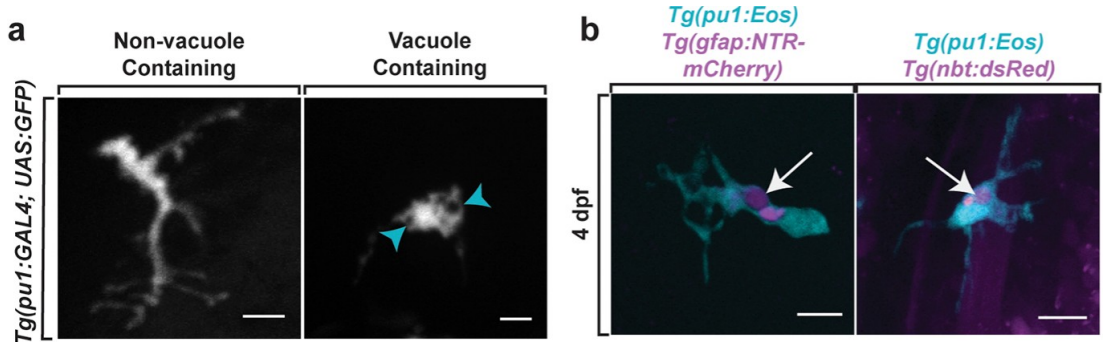
Gordon et al. (2024), pbio.3002819.s008.avi. Academic: Microglial cell dynamics.

Phagocytic vacuoles identify a cell's recent activity

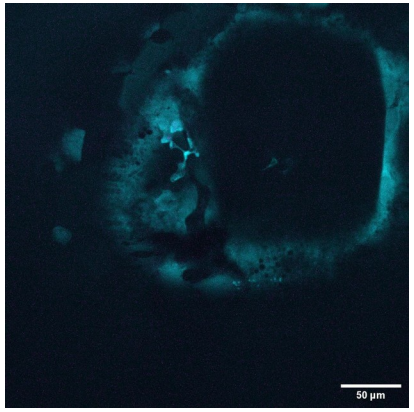
The zebrafish study followed microglia containing **vacuoles**, internal compartments that held engulfed material. Labels in neurons and other glial cells identified the origins of some debris. Researchers could track what microglia had taken up rather than inferring their activity from their location alone.

Microglia that became phagocytic frequently died during the subsequent developmental observation period. Their internal contents therefore carried information about their recent behavior and later fate. The investigators followed individual cells, making it possible to relate engulfment and survival within the same living tissue.

The result concerns embryonic zebrafish microglia during an active period of brain construction. Cell identity alone was insufficient to predict a long lifespan. The cell's recent phagocytic history also affected its fate, adding a dynamic cellular process to the maintenance of developing nervous tissue.



Gordon et al. (2024), Fig. 2a–b. Microglial turnover and debris handling.



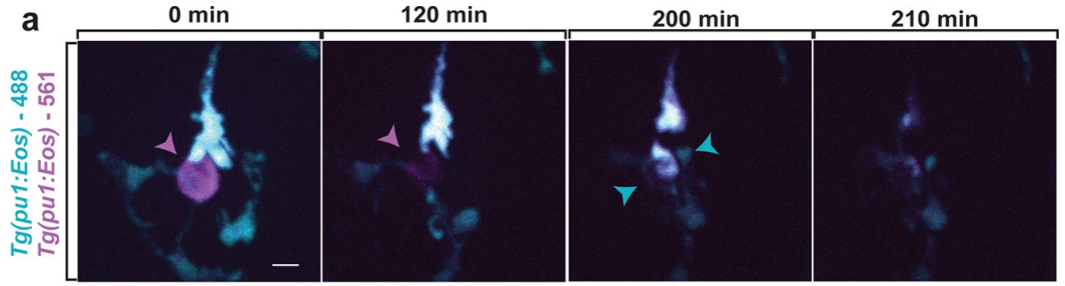
Gordon et al. (2024), pbio.3002819.s009.avi. Academic: Microglial cell dynamics.

Microglia can engulf debris from other microglia

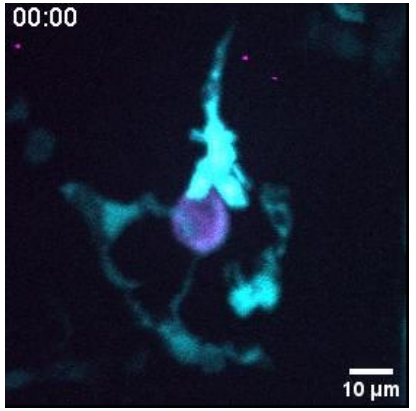
The researchers used a photoconvertible fluorescent protein to distinguish selected microglia from their neighbors. **Photoconversion** changes a marker's fluorescence after illumination. Following the converted label allowed them to determine whether material from one microglial cell later appeared inside a different microglial cell.

After labeled microglia died, neighboring microglia engulfed their debris. Material from one resident immune cell therefore passed into another. The authors described this as microglial cannibalism. The process removed fragments but also created a new phagocytic cell whose later survival could be followed.

The investigators proposed a cycle in which engulfment, cell death, and removal of microglial debris produce further engulfment. The cycle concerns the developmental cell population they measured. It explains how tissue clearance can itself generate additional debris when the cells performing the clearance subsequently die.



Gordon et al. (2024), Fig. 3a. Microglial turnover and debris handling.



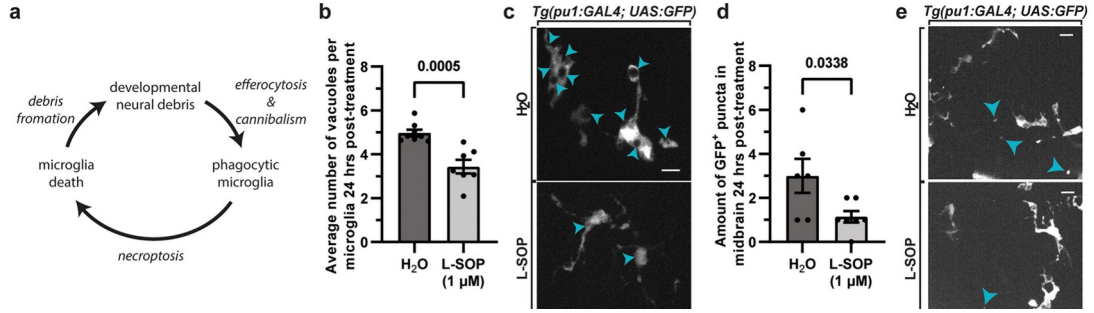
Gordon et al. (2024), S5 MovieMicroglia cannibalism. Academic: Microglial cell dynamics.

Phagocytosis changes developmental cell survival

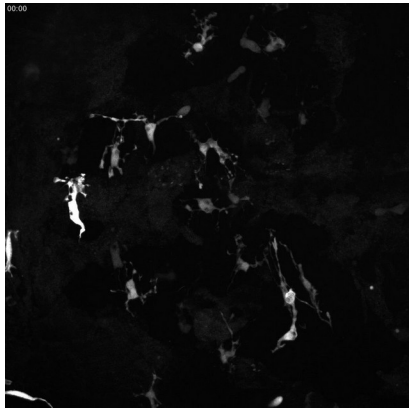
The investigators altered the amount of neural debris and manipulated phagocytosis in developing zebrafish. These experiments tested whether engulfment contributed to microglial death. Following cells after a manipulation provided evidence about the process, beyond simply observing that vacuoles and later death occurred together.

Reducing phagocytic activity reduced the burden of internal debris and microglial loss under the study conditions. Increasing neural debris drove more engulfment. The cells' surroundings therefore changed their behavior and survival, even though they retained the same identity as resident microglia in the developing brain.

The study describes maintenance as a process with costs for the cells performing it. Engulfing unwanted material can protect tissue while changing the fate of the phagocyte. Neuronal survival and function depend on this broader cellular environment, rather than on the electrical properties of individual neurons alone.



Gordon et al. (2024), Fig. 4a–e. Microglial turnover and debris handling.



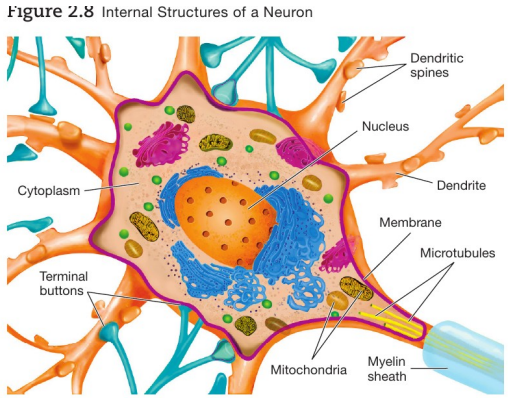
Gordon et al. (2024), pbio.3002819.s011.avi. Academic: Microglial cell dynamics.

Cell maintenance operates on a different timescale

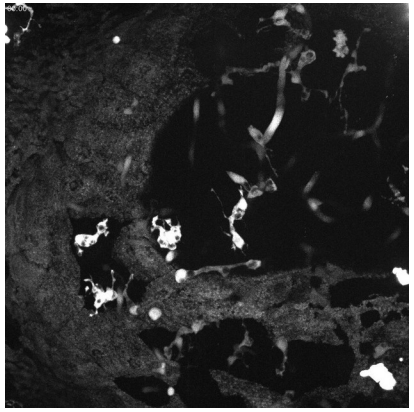
A neuron maintains its membrane, internal proteins, and organelles throughout its life. Surrounding glial cells also maintain the tissue. These activities occur continuously, even during periods when a neuron is not firing. A quiet electrical recording therefore does not mean that the cell or its surroundings are biologically inactive.

Protein synthesis, intracellular transport, and debris removal take longer than the electrical events used for rapid communication. The processes have different physical mechanisms. A membrane current moves ions across a membrane, while a moving cell changes its shape and position through coordinated activity of its internal structural proteins.

Time-lapse microscopy provides visual context for this slower side of nervous-system function. Fluorescent labels identify cellular structures while successive frames reveal their movement. These recordings concern maintenance and cellular behavior; they should not be read as direct measurements of the fast voltage changes that generate action potentials.



Carlson and Birkett (2022), Fig. 2.8, p. 42.



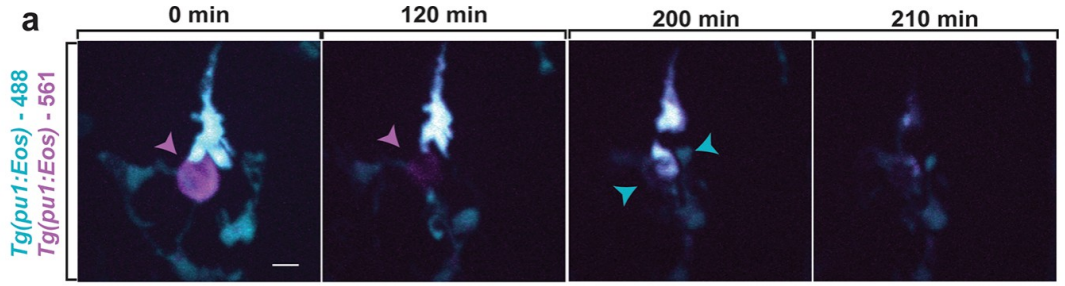
Gordon et al. (2024), pbio.3002819.s012.avi. Aesthetic context: Microglial cell dynamics.

Phagocytosis maintains the neural environment

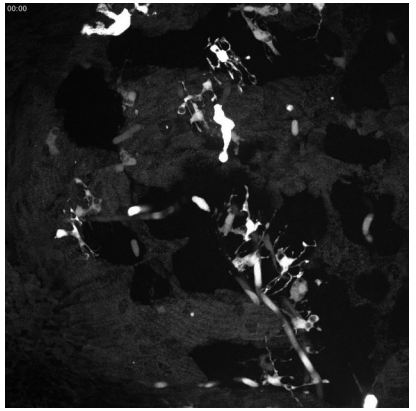
Removing dead cells and fragments helps maintain the environment surrounding surviving neurons. Microglia take up this material into internal compartments, where it can be processed. The activity changes which cellular remnants remain in the tissue and which cells contain the material after it has been engulfed.

Debris handling is different from the ionic regulation performed by other glial mechanisms. The nervous system requires both forms of maintenance. Neuronal electrical signals depend on a regulated extracellular fluid, while the living tissue also needs to remove material produced by cellular turnover, damage, and development.

Phagocytosis brings extracellular material across the boundary of an immune cell. A fluorescent fragment may therefore move from one cell into another without representing a newly formed neuron or synapse. Tracking the identity of the surrounding cell helps distinguish debris uptake from growth of a new cellular structure.



Gordon et al. (2024), Fig. 3a. Microglial turnover and debris handling.



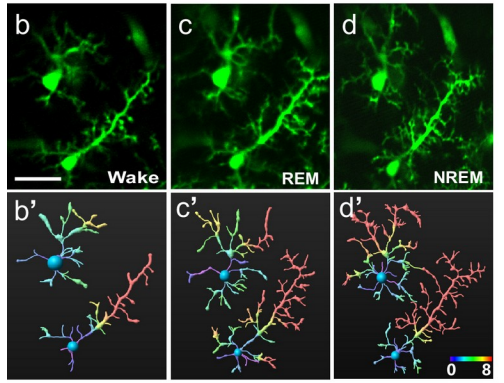
Gordon et al. (2024), pbio.3002819.s013.avi. Aesthetic context: Microglial cell dynamics.

Microglial surveillance changes with brain state

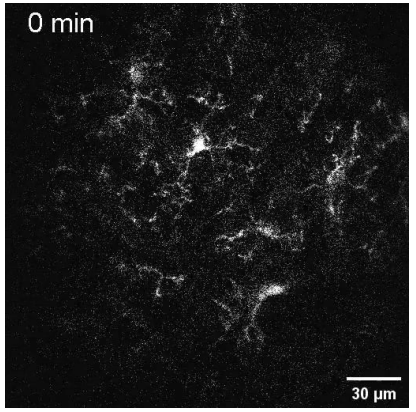
A study in freely behaving mice used a miniature two-photon microscope to follow microglia in the somatosensory cortex. Researchers also recorded brain electrical activity, muscle activity, and behavior. These measurements allowed them to identify sleep and waking while observing the same resident immune cells.

Microglial surveillance changed with sleep–wake state, and sleep deprivation also changed the cells' responses. The investigators measured signaling by **norepinephrine**, a neurotransmitter, and manipulated the pathway supplying it. A chemical signal associated with brain state therefore influenced the behavior of cells responsible for local tissue maintenance.

The **locus coeruleus** is a brainstem nucleus that supplies norepinephrine. The study identified contributions from this pathway and **beta-2 adrenergic receptors**, which respond to norepinephrine. Norepinephrine signaling therefore changes how immune cells perform maintenance within the brain as the surrounding neural state changes.



Gu et al. (2023), Fig. 2b–d. Microglial surveillance across sleep and waking.



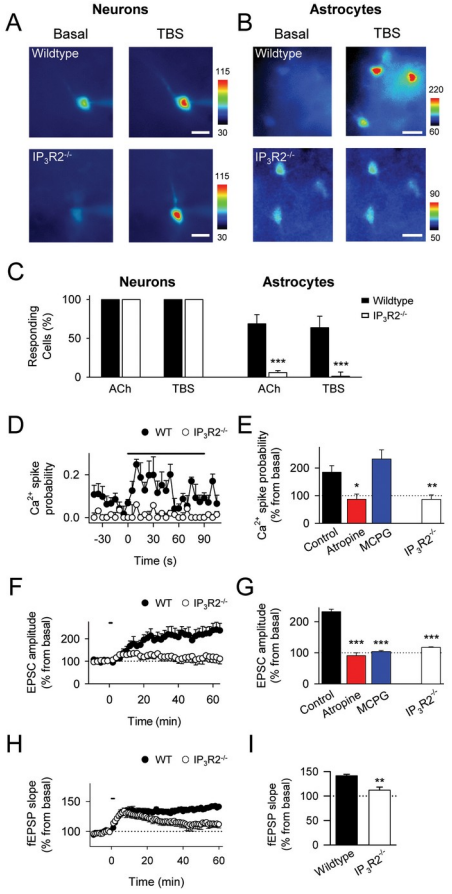
Gu et al. (2023), Figure 1, video 1. Aesthetic context: Microglial surveillance.

Astrocytic calcium contributes to synaptic strengthening

The investigators reduced calcium signaling in astrocytes using intracellular interventions and mice lacking the receptor needed for a major internal calcium-release pathway. They then tested whether cholinergic stimulation still strengthened transmission. **Long-term potentiation**, or **LTP**, is a persistent increase in the effectiveness of a synapse.

Suppressing the relevant astrocytic calcium signal prevented cholinergically induced potentiation in these experiments. Restoring a calcium rise while depolarizing the postsynaptic neuron restored the effect. The timing of astrocytic and neuronal activity therefore contributed to whether the synapse underwent a lasting increase in transmission.

The results identify a particular mechanism under the tested conditions. Cholinergic input recruited astrocytic signaling, which helped engage glutamate-sensitive metabotropic receptors during potentiation. The resulting change increased the effectiveness of later transmission through the tested pathway, giving the timing of glial and neuronal events a role in the persistent synaptic response.



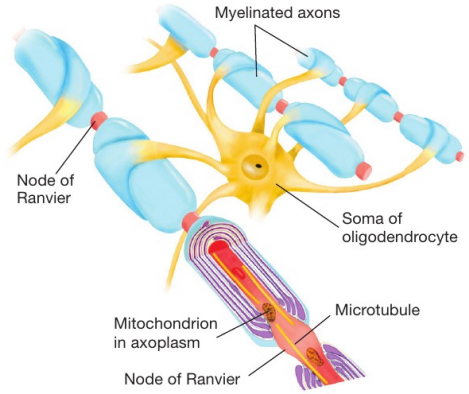
Navarrete et al. (2012), Fig. 4. Astrocytic calcium and hippocampal potentiation.

Oligodendrocytes wrap axons with myelin membrane

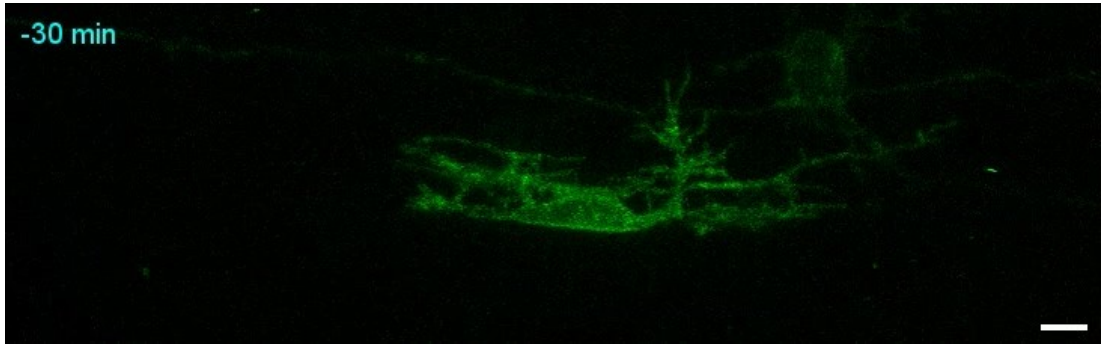
Myelin is a layered membrane sheath around an axon. In the central nervous system, an **oligodendrocyte** forms myelin around segments of axons. The wrapping changes the electrical properties of the axonal membrane and helps signals travel efficiently between the exposed gaps along a myelinated fiber.

Myelin is made by a separate cell rather than by the axon itself. An oligodendrocyte extends processes that contact axons and form membrane sheaths. One oligodendrocyte can contribute sheaths to several axonal segments, so its condition can affect the function of more than one part of a circuit.

A living sheath begins as a contact between a glial process and an axon. Time-lapse imaging records how these contacts develop, expand, or disappear. The developmental movement of membrane should be distinguished from the rapid electrical conduction that an established sheath later helps support along the axon.



Carlson and Birkett (2022), Fig. 2.11, p. 45.



Almeida et al. (2023), Figure 9, video 2. Aesthetic context: Oligodendrocyte ensheathment.

Schwann cells myelinate peripheral axonal segments

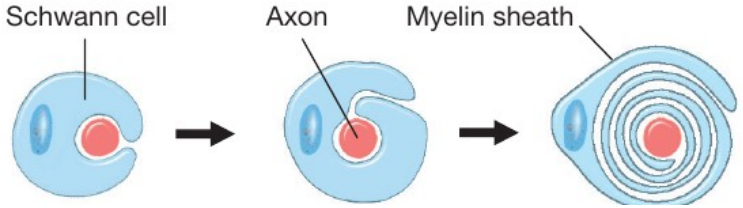
A **Schwann cell** forms myelin in the peripheral nervous system. Each myelinating Schwann cell wraps one segment of one axon. This differs from a central oligodendrocyte, which can extend several processes and form multiple sheaths. Both cell types place insulating membrane around axonal segments.

Gaps between adjacent myelinated segments are called **nodes of Ranvier**. The exposed membrane at these gaps contains channels that participate in regenerating action potentials. The sheath and the node therefore perform different jobs: the sheath alters passive current spread, while the node provides membrane that actively regenerates the electrical signal.

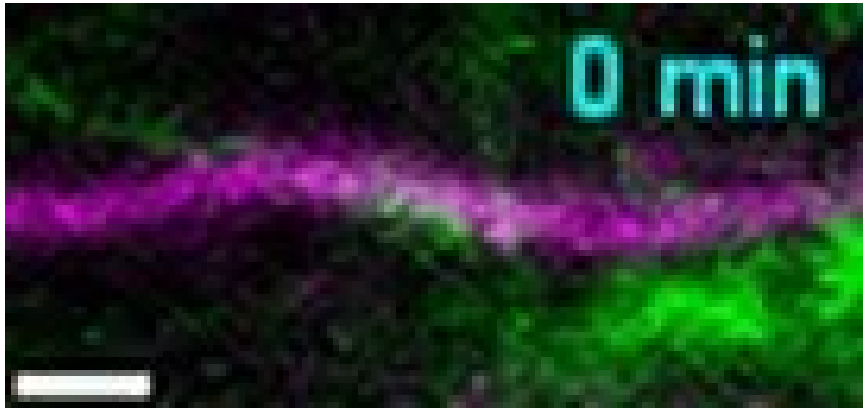
Central and peripheral myelin have different cellular sources, but both require an interaction between an axon and a supporting cell. A peripheral sheath develops through the wrapping of one Schwann cell, whereas central sheath development involves an oligodendrocyte process. This cellular distinction affects how the axonal covering is constructed and maintained.

Figure 2.12 Formation of Myelin

In the peripheral nervous system, an entire Schwann cell tightly wraps itself many times around an individual axon and forms one segment of the myelin sheath.



Carlson and Birkett (2022), Fig. 2.12, p. 45.



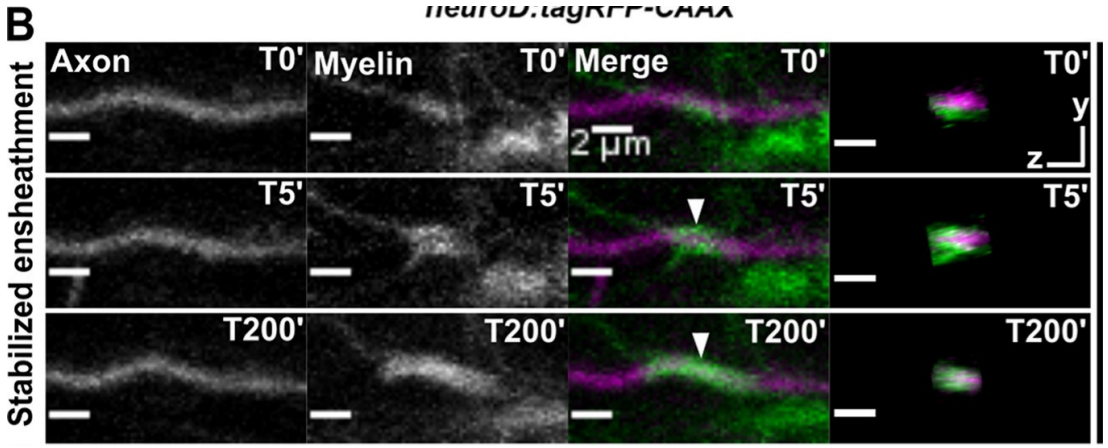
Almeida et al. (2023), Figure 2, video 1. Aesthetic context: Oligodendrocyte ensheathment.

Early axonal wrapping is repeatedly started and lost

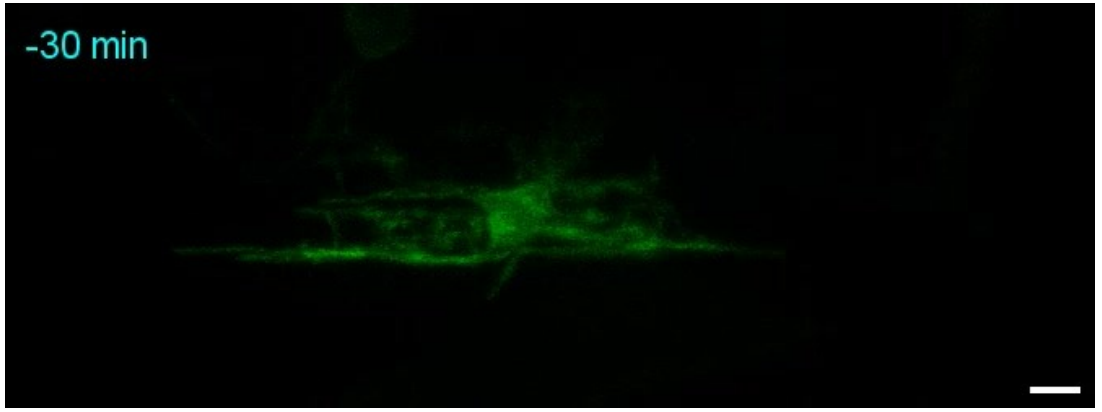
A 2023 study followed individual oligodendrocytes in the developing zebrafish spinal cord. Longitudinal imaging allowed researchers to revisit the same cells and axons. They counted initial ensheathments, their disappearance, and the sheaths that remained, instead of measuring only the final number of visible sheaths.

Oligodendrocytes repeatedly contacted and ensheathed the same axons before stable sheaths formed. Many initial ensheathments disappeared. The number of mature sheaths therefore reflected both the initiation of new contacts and the persistence of earlier contacts. A contact observed at one moment was not necessarily a lasting sheath.

The result identifies myelination as a dynamic developmental process. A cell's eventual sheath pattern cannot be inferred solely from one early image. Repeated observations distinguish transient membrane interactions from stable wrapping and explain how a developing oligodendrocyte can generate a very different final pattern from its initial contacts.



Almeida et al. (2023), Fig. 2B. Oligodendrocyte ensheathment dynamics.



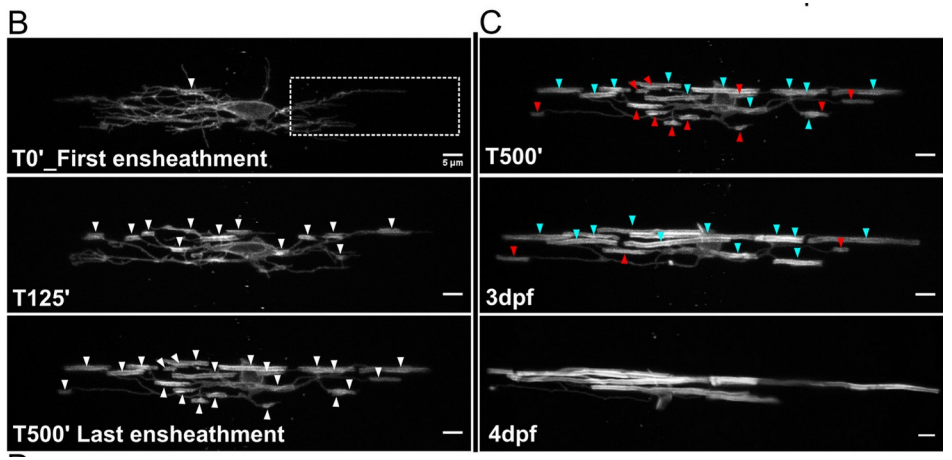
Almeida et al. (2023), Figure 6, video 2. Academic: Oligodendrocyte ensheathment.

Sheath stabilization differs from sheath initiation

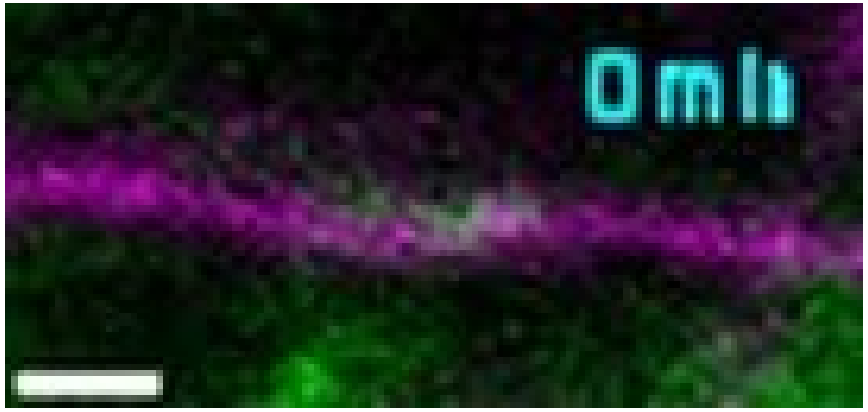
The investigators separated the first appearance of an ensheathment from its later stabilization. Some oligodendrocytes initiated many contacts, while others initiated fewer. Following the same cells over time made it possible to ask whether cells differed because of how often they started wrapping or how reliably wraps persisted.

Most early ensheathments disappeared, and the loss fraction was relatively consistent across cells. Differences in the total number of attempts helped generate differences in final sheath number. Cells could therefore reach different outcomes even when a broadly similar proportion of their early contacts failed to persist.

The study also examined membrane recycling through the protein **Rab5**. Disrupting this pathway increased later sheath loss without producing the same change in early initiation. Membrane handling thus affected stabilization differently from the first contact, separating two cellular steps that a final myelin image would combine.



Almeida et al. (2023), Fig. 4B–C. Oligodendrocyte ensheathment dynamics.



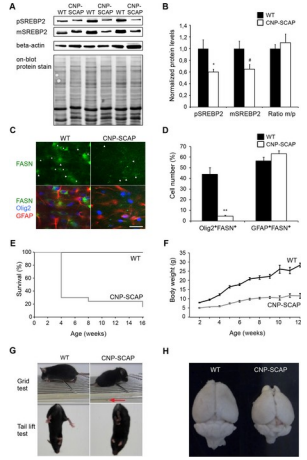
Almeida et al. (2023), Figure 2, video 3. Academic: Oligodendrocyte ensheathment.

Astrocytes supply lipids used to build central myelin

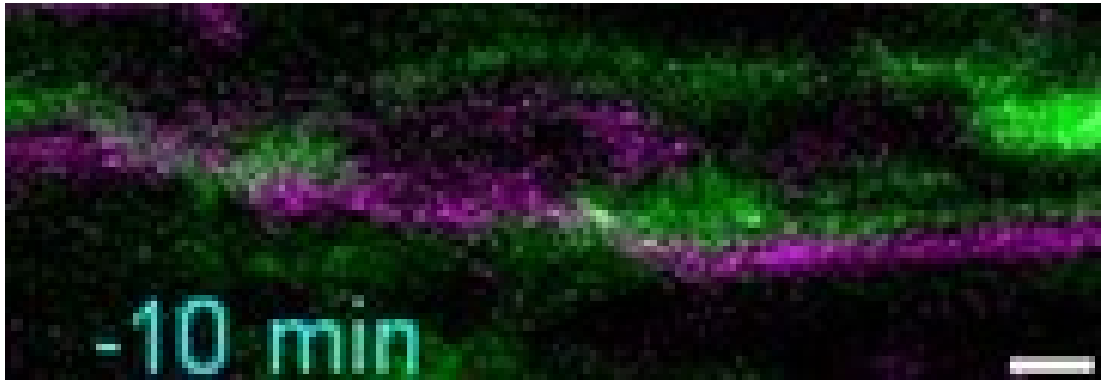
A separate mouse study tested where oligodendrocytes obtain the lipids needed for myelin. **Lipids** are major membrane components. Researchers disrupted a regulator of lipid synthesis in selected glial populations, allowing them to compare the contributions of oligodendrocytes and astrocytes to central myelin formation.

Disrupting lipid synthesis in oligodendrocytes delayed myelination, while disrupting it in astrocytes produced persistent myelin deficits. The myelinating cell therefore depended partly on material supplied by a neighboring glial population. Myelin construction required metabolic cooperation across cell types, in addition to the oligodendrocyte's own synthetic machinery.

The experiment identifies the supply of membrane-building material as a distinct part of the mechanism. A glial process can contact an axon yet still require adequate resources to build a sheath. The glial cell must acquire these resources while extending and stabilizing its processes along axons. The metabolic supply and the physical wrapping process are distinct requirements for making a functional sheath.



Camargo et al. (2017), Fig. 1. Glial lipid supply and central myelin.



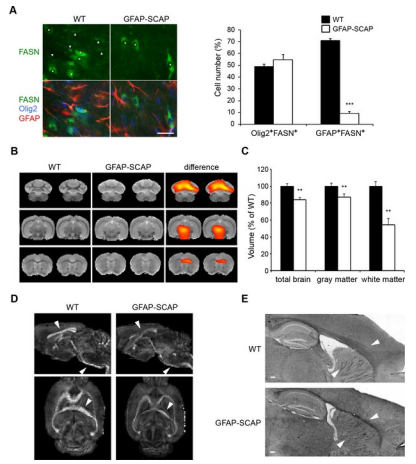
Almeida et al. (2023), Figure 3, video 1. Aesthetic context: Oligodendrocyte ensheathment.

Extracellular lipid supply can alter myelin formation

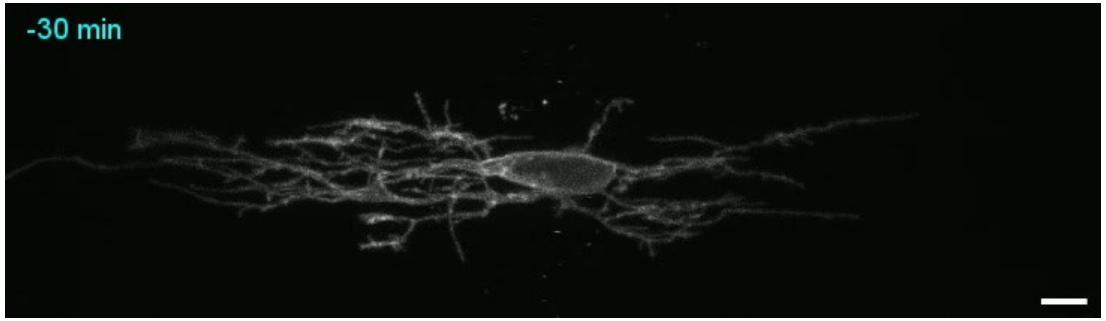
When astrocytic lipid synthesis was disrupted, oligodendrocytes incorporated more circulating lipids into myelin. Researchers also tested a lipid-enriched diet in the affected mice. These experiments asked whether extracellular material could compensate for a deficit in the local supply of components used to build the sheath.

The enriched diet rescued hypomyelination in the conditional mutants studied. **Hypomyelination** means that less myelin forms than expected. The result identifies a source of membrane material that can influence central myelination when a particular glial lipid-synthesis pathway is impaired in this experimental preparation.

The mechanism involves obtaining building materials, not simply commanding an axon to fire more often. Lipid availability and membrane stabilization affect different steps of sheath production. Distinguishing those steps explains why neural conduction depends on both the axon's electrical machinery and the cells that construct its surrounding insulation.



Camargo et al. (2017), Fig. 4. Glial lipid supply and central myelin.



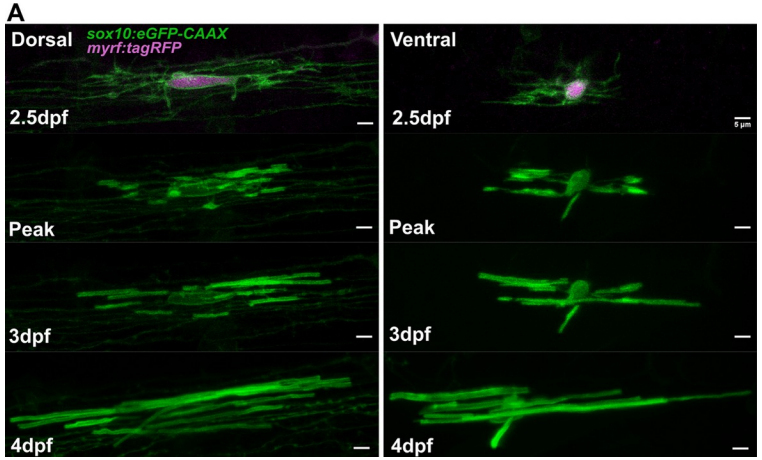
Almeida et al. (2023), Figure 4, video 1. Aesthetic context: Oligodendrocyte ensheathment.

Glial membrane handling shapes the final sheath pattern

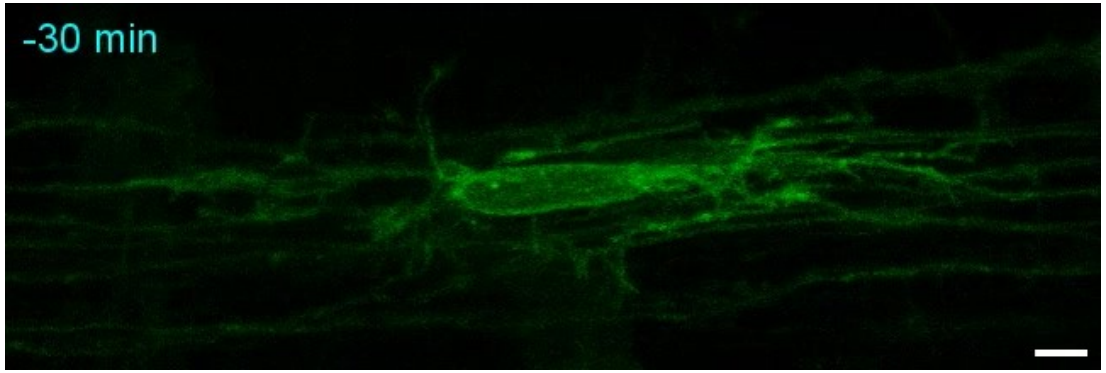
The zebrafish investigators compared ensheathment dynamics in different regions and developmental periods. They tracked individual glial processes rather than treating all wrapping events as one average structure. The observations captured repeated attempts, loss, and persistence as a living oligodendrocyte established its collection of myelin sheaths.

Their analyses separated variation in contact initiation from a relatively consistent rate of stabilization. The final pattern therefore emerged from a sequence of cellular events. A cell that started more ensheathments could finish with more stable sheaths even while losing a large share of its initial attempts.

The Rab5 experiments further indicated that membrane recycling contributes to maintaining later sheaths. The authors distinguished this stabilization phase from early initiation. Myelin organization therefore depends on how membrane is handled over time, providing a cellular foundation for the insulation that supports reliable axonal conduction.



Almeida et al. (2023), Fig. 6A. Oligodendrocyte ensheathment dynamics.



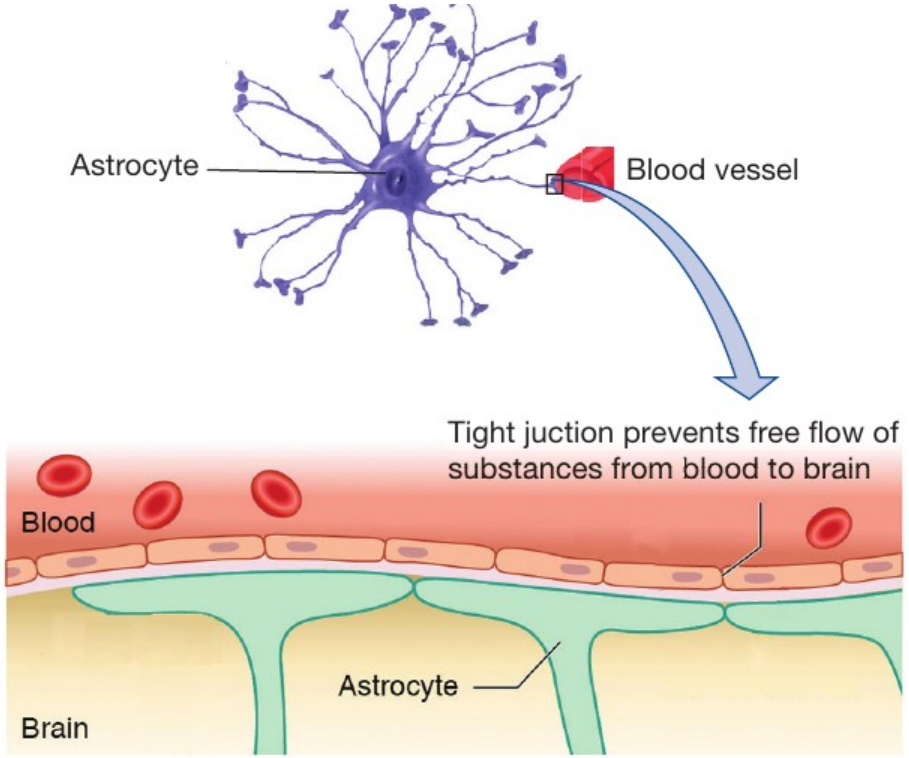
Almeida et al. (2023), Figure 6, video 1. Aesthetic context: Oligodendrocyte ensheathment.

The blood–brain barrier controls the neural environment

The **blood–brain barrier** limits movement of many substances from circulating blood into brain tissue. Cells lining brain capillaries form tight junctions that restrict passage between them. Materials entering the brain must therefore often cross the lining cells through particular transport pathways rather than leaking freely between neighboring cells.

This selective boundary helps maintain the extracellular conditions needed for neuronal communication. Blood composition changes as the body eats, exercises, or encounters chemicals. Brain cells require a more regulated environment because their signaling depends on ion concentrations and on chemical messengers acting at appropriate locations and times.

The barrier is selective rather than completely impermeable. Some molecules cross readily, while others require transporters or enter poorly. This distinction affects both normal brain nutrition and the ability of a drug in the bloodstream to reach its target within the central nervous system.



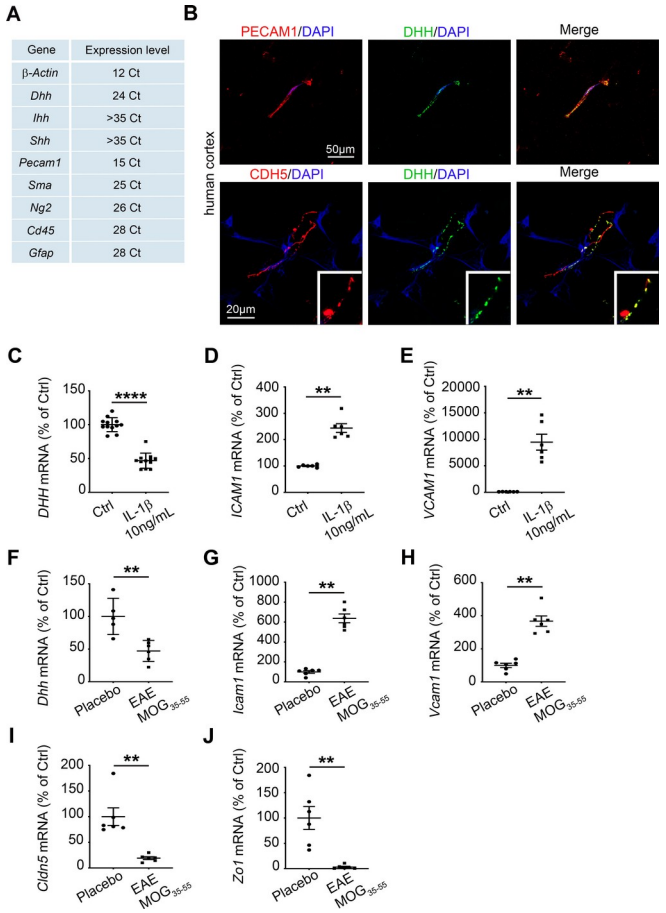
Carlson and Birkett (2022), Fig. 2.13, p. 46.

Endothelial signals help maintain a tight brain barrier

A mouse study examined **desert hedgehog**, a signaling protein expressed by endothelial cells at the blood–brain barrier. Endothelial cells form the inner lining of blood vessels. Researchers selectively disrupted this signal and measured how the barrier and surrounding glial tissue responded to the change.

Disrupting endothelial desert hedgehog compromised the vascular barrier under the experimental conditions. Its role therefore extended beyond providing a physical cell layer. Communication among endothelial cells helped maintain the properties that normally restrict exchange between the blood and the tissue containing neurons and glia.

The experiment separates a barrier's cellular structure from the signals maintaining that structure. A capillary can remain physically present while its permeability changes. This distinction helps explain why normal-looking anatomy alone is insufficient to establish that the surrounding extracellular environment is still regulated in the same way.



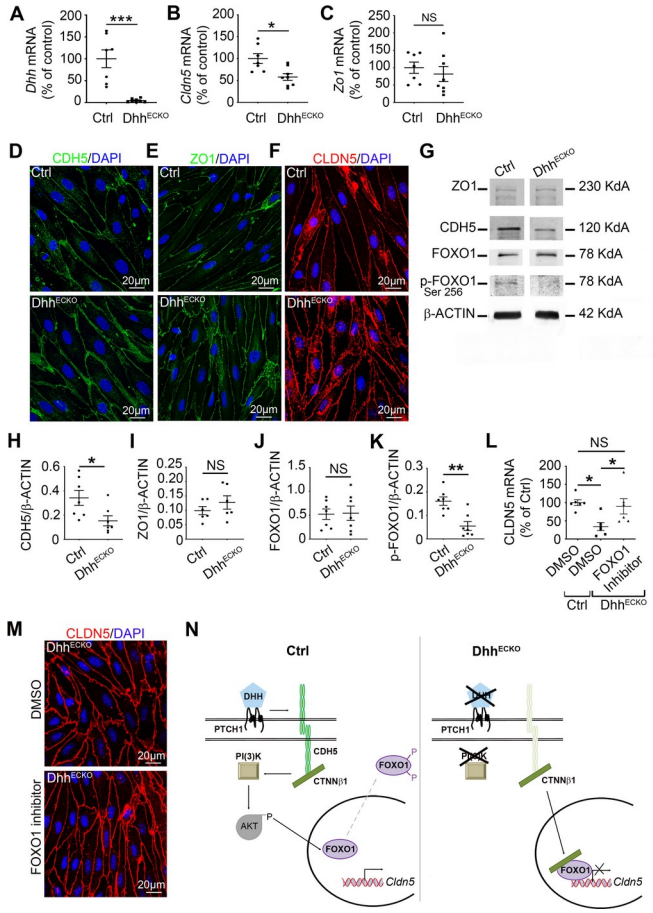
Mora et al. (2020), Fig. 1. Endothelial and glial barrier responses.

Astrocytes can form a second protective boundary

The investigators examined the **glia limitans**, a boundary associated with astrocytic processes around the central nervous system and its vessels. After endothelial barrier disruption, they found changes in astrocytes that included formation of a tighter secondary boundary. Different cellular layers therefore responded differently to the same manipulation.

The new glial boundary restricted entry into the neural tissue even though the endothelial barrier had been compromised. Protection depended on a response by nearby cells, rather than on immediate restoration of the original vascular barrier. The tissue reorganized its boundaries after the endothelial signaling change.

The authors described a reciprocally regulated double-barrier system. Endothelial and astrocytic barriers were distinct structures with interacting responses. Their findings emphasize that a neural environment is maintained by coordinated activity across cell types, with a second boundary sometimes compensating for a change in the first.



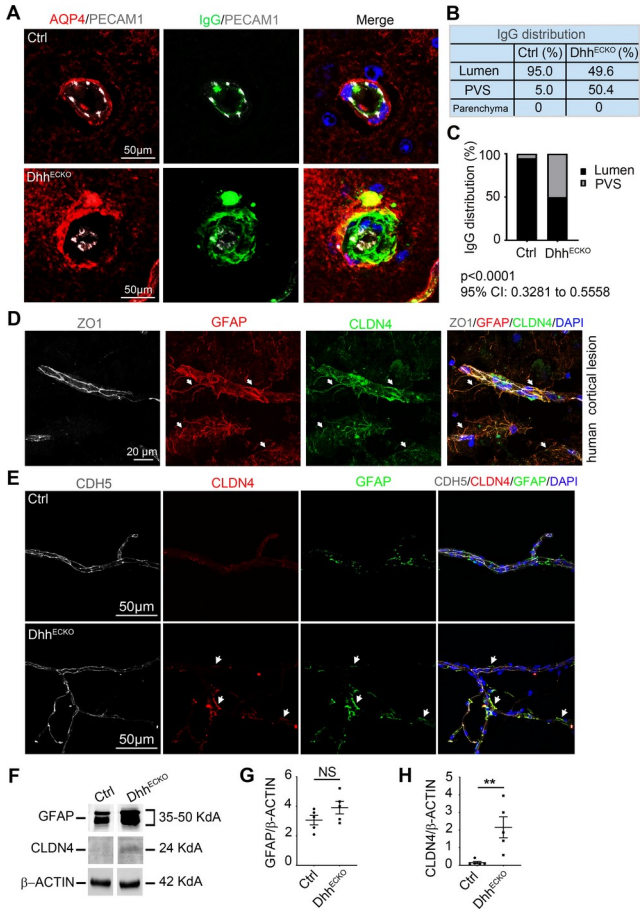
Mora et al. (2020), Fig. 2. Endothelial and glial barrier responses.

Barrier changes alter later inflammatory responses

The study also tested how prior barrier disruption affected subsequent experimental inflammation. Researchers compared mice with the endothelial signaling manipulation to appropriate controls. They examined the secondary glial boundary and the later response of the central nervous system when an inflammatory challenge was introduced.

The induced glial boundary limited the severity of later inflammation in the model. The authors proposed the term **chronic neuroinflammatory tolerance** for this protective state. The outcome depended on the sequence of barrier disruption, tissue response, and later challenge, rather than on a single unchanged vascular structure.

These results explain how the same general process, altered barrier permeability, can produce different outcomes depending on the tissue's response. Endothelial cells, astrocytes, and inflammatory cells act in a coordinated environment. Their interactions determine which substances and cells reach the neural tissue after a barrier change.



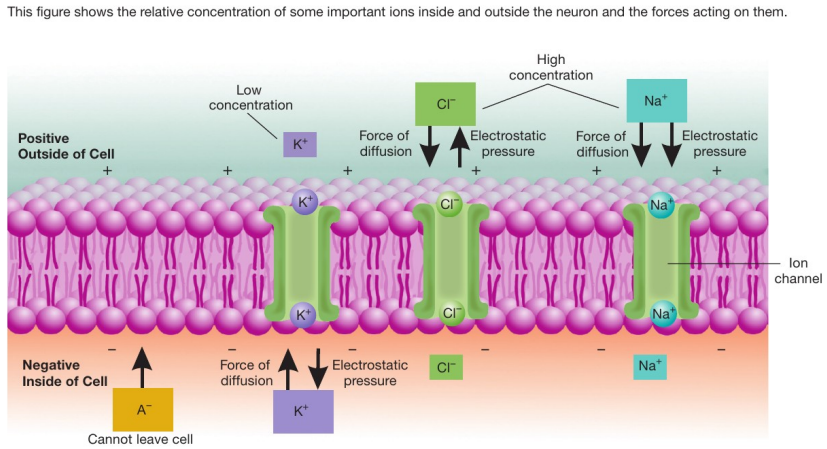
Mora et al. (2020), Fig. 4. Endothelial and glial barrier responses.

Membranes maintain unequal ion concentrations

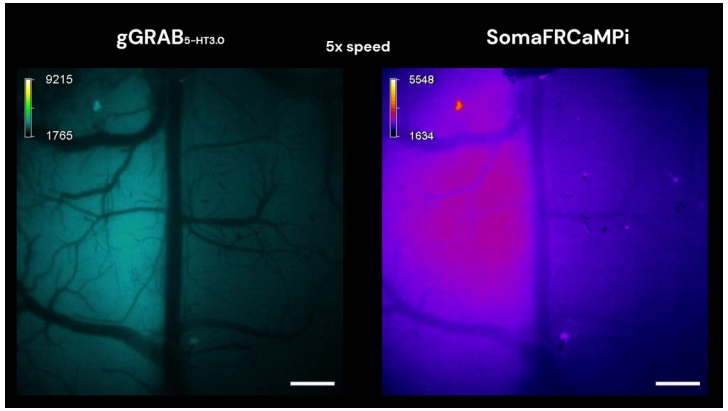
The neuronal membrane separates intracellular fluid from extracellular fluid. Both contain **ions**, atoms or molecules carrying electrical charge. Sodium, potassium, chloride, and calcium have different concentrations on the two sides. These differences provide conditions that ion channels can use to change electrical activity rapidly.

A **concentration gradient** favors movement from a region of higher concentration toward a region of lower concentration. Electrical forces also affect charged particles. An ion's movement through an open channel depends on both influences, so concentration alone is insufficient to determine the direction of its net movement.

The membrane's lipid layer prevents many charged particles from crossing freely. Selective protein channels provide routes through it. The cell can therefore maintain unequal concentrations while permitting particular ions to move when the appropriate channels open, converting stored ionic differences into changing electrical signals.



Carlson and Birkett (2022), Fig. 2.17, p. 51.



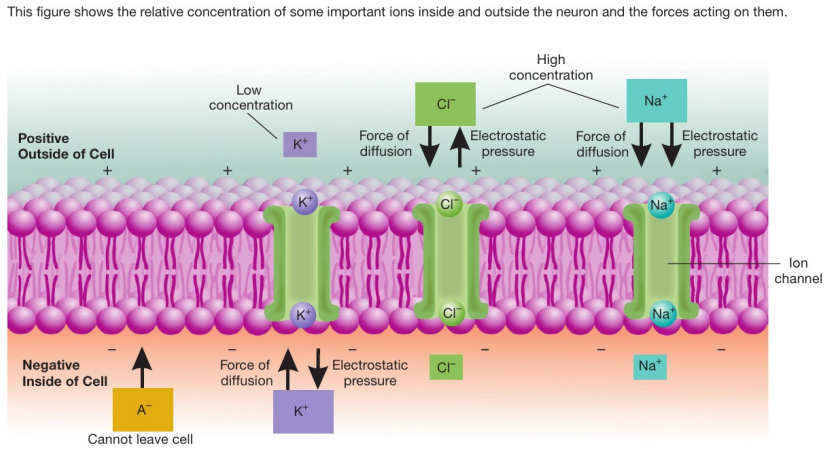
Zhou et al. (2025), pbio.3003048.s013.mp4. Aesthetic context: Cellular calcium signals.

Membrane voltage measures charge separation

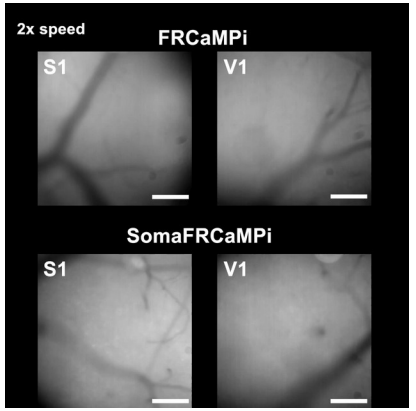
The **membrane potential** is the voltage difference between the inside and outside of a cell. A recording electrode inside an axon can be compared with an electrode outside it. At rest, the inside of a typical neuron is electrically negative relative to the surrounding fluid.

Only a small separation of charge near the membrane is needed to produce this voltage. The fluids on both sides still contain many positive and negative ions. Membrane voltage therefore describes an electrical difference across a thin boundary, rather than the entire interior becoming a reservoir of one charge.

A **depolarization** makes the membrane potential less negative. A **hyperpolarization** makes it more negative. These terms describe changes relative to an earlier potential. Neither word by itself identifies which ion moved; the relevant channels and ionic gradients determine how that voltage change was produced.



Carlson and Birkett (2022), Fig. 2.17, p. 51.



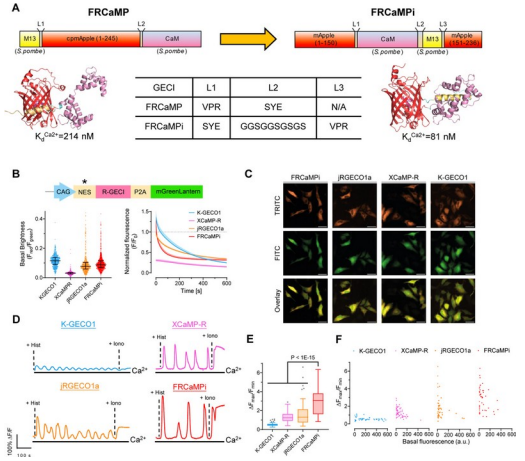
Zhou et al. (2025), pbio.3003048.s016.mp4. Aesthetic context: Cellular calcium signals.

Fluorescent calcium sensors report intracellular chemistry

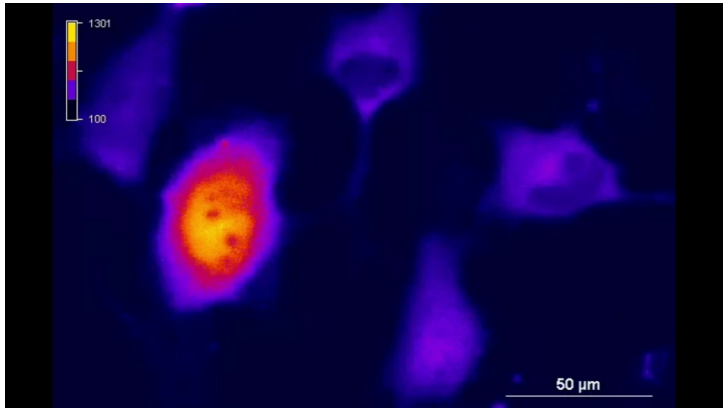
A 2025 study developed a red fluorescent calcium indicator called **FRCaMPi** and a soma-localized version. An indicator changes its fluorescence when calcium binds to it. Researchers first characterized the sensor in cells and then tested recordings from neuronal populations in living animals.

The fluorescence signal reports changes in calcium associated with cellular activity. Calcium can rise when neurons are active, but it is a different physical variable from membrane voltage. A sensor's response also takes time, so the fluorescent trace has its own sensitivity and kinetics.

Restricting a sensor toward the soma helps distinguish signals from neighboring cells whose processes overlap. The study compared indicator variants in several preparations. Those comparisons tested whether a particular molecular design could improve the ability to assign an observed activity signal to the cell that produced it.



Zhou et al. (2025), Fig. 1. Fluorescent calcium indicators and cellular responses.



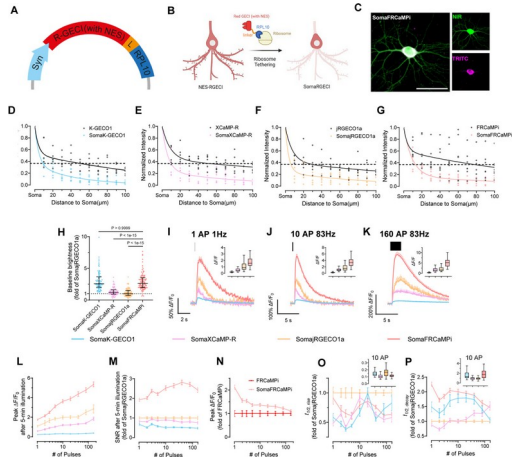
Zhou et al. (2025), pbio.3003048.s012.mp4. Academic: Cellular calcium signals.

Visual stimuli recruit calcium responses in fish neurons

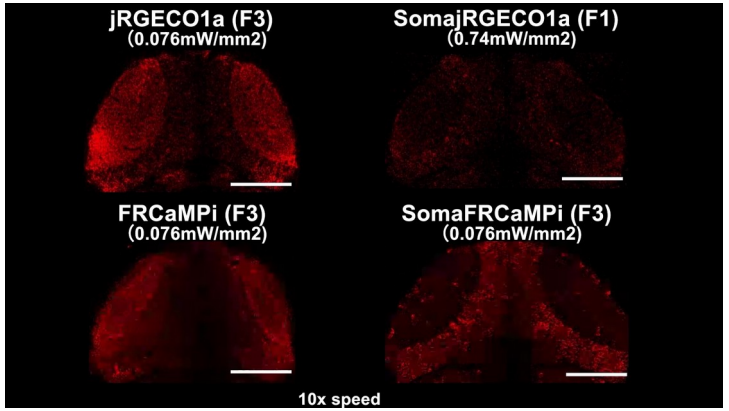
The sensor study imaged the optic tectum of larval zebrafish during looming visual stimuli. A **looming stimulus** expands in the visual field, resembling an approaching object. Researchers compared fluorescent indicators while neurons responded to this defined sensory event in an intact animal.

The preparation allowed calcium responses to be measured from populations of cells during visual stimulation. Sensor localization and sensitivity affected how individual responses could be resolved. The experiment therefore addressed both a biological signal, stimulus-related neuronal activity, and the performance of the molecular tool used to record it.

The fluorescence traces should be interpreted as calcium measurements rather than direct counts of every action potential. The indicator binds calcium and changes brightness according to its own response properties. The study's comparisons provide the evidence needed to judge how well the resulting signal separates activity in individual neurons.



Zhou et al. (2025), Fig. 3. Fluorescent calcium indicators and cellular responses.



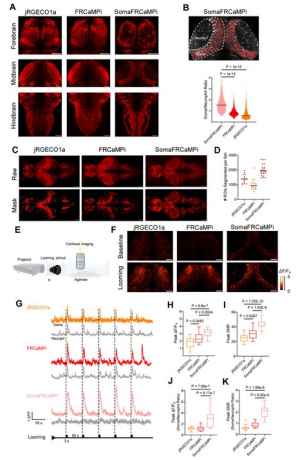
Zhou et al. (2025), pbio.3003048.s014.mp4. Academic: Cellular calcium signals.

Calcium imaging can resolve activity in awake cortex

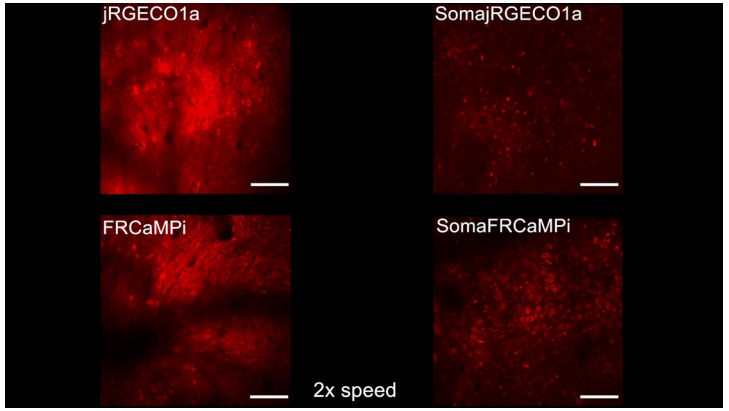
The investigators also recorded neurons in the primary visual cortex of awake mice. Two-photon imaging allowed fluorescence to be measured from cells within the tissue. They compared the red indicator and soma-localized variants to determine how the sensors performed under these recording conditions.

Different neurons produced distinguishable changes in fluorescence over time. Confining more of the signal to the soma reduced contributions from overlapping processes. This molecular targeting helped assign changing brightness to individual cells rather than treating all fluorescence within the field as activity from one common source.

The method makes neuronal activity accessible optically, while retaining the distinction between calcium and voltage. The sensor does not replace the neuron's ion channels; it reports one consequence of cellular calcium changes. Interpreting its trace therefore requires both the biological preparation and the indicator's measured response properties.



Zhou et al. (2025), Fig. 4. Fluorescent calcium indicators and cellular responses.



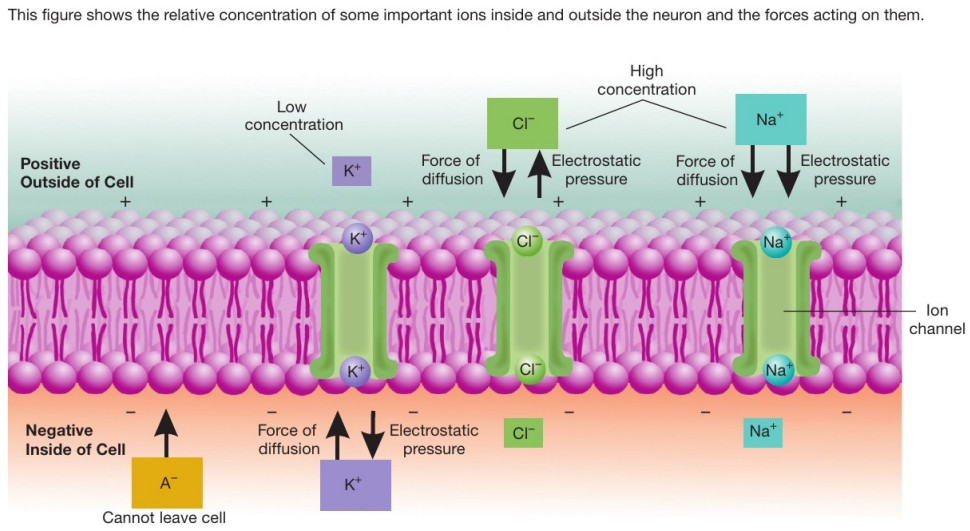
Zhou et al. (2025), pbio.3003048.s015.mp4. Academic: Cellular calcium signals.

Selective permeability helps establish the resting state

At rest, a neuronal membrane is more permeable to some ions than others. Potassium can move through resting channels, while many sodium channels are closed. The resulting voltage depends on which routes are available and on the concentration and electrical forces acting on each ion.

Potassium tends to move outward down its concentration gradient. As positive charge leaves, the cell interior becomes more negative, creating an electrical influence that opposes further outward movement. The resting potential reflects the combined effects of several ions rather than the concentration gradient of one substance alone.

Opening an additional channel changes the membrane's permeability. The effect on voltage depends on which ion the channel conducts and the forces acting on it at that moment. A channel's identity and the existing membrane potential therefore determine whether opening it promotes depolarization or opposes that change.



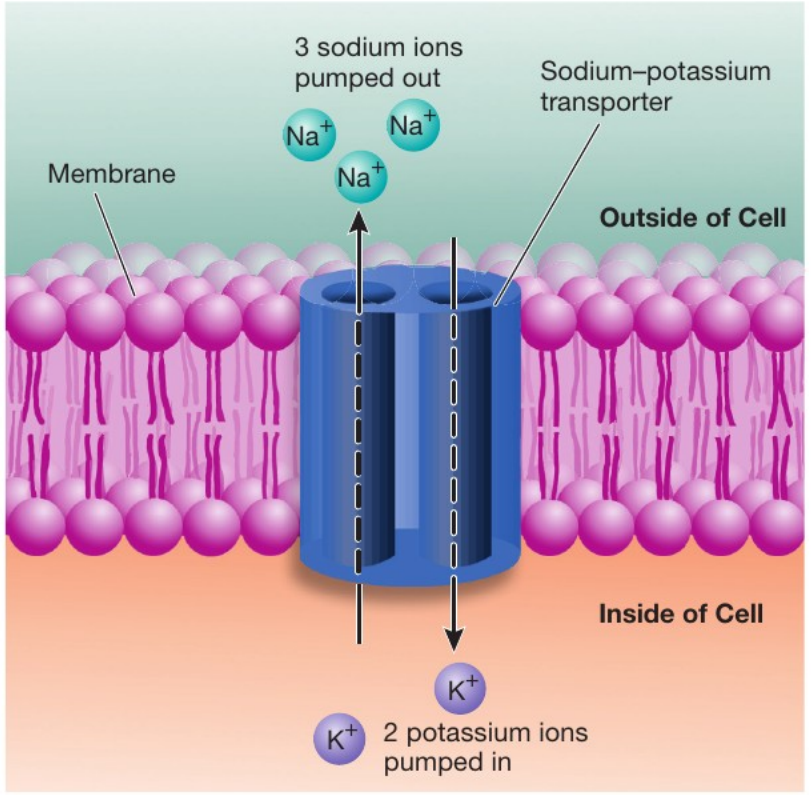
Carlson and Birkett (2022), Fig. 2.17, p. 51.

Ion pumps maintain sodium and potassium gradients

The **sodium–potassium pump** uses energy from ATP to move sodium out of the cell and potassium into it. This transport works against their concentration gradients. Its activity helps maintain the unequal ion concentrations that the neuron repeatedly uses when channels open during signaling.

The pump and an ion channel perform different tasks. A channel provides a selective route through which an ion moves according to existing forces. The pump consumes metabolic energy to maintain the concentration differences. A rapid action potential results from changing channel activity, not from a sudden pump-driven electrical discharge.

Repeated signaling changes the distribution of ions across the membrane. Maintaining the gradients therefore requires continuing metabolic work, even though a single action potential transfers only a small fraction of the available ions. Cellular energy supply and membrane signaling operate together across different timescales in the same neuron.



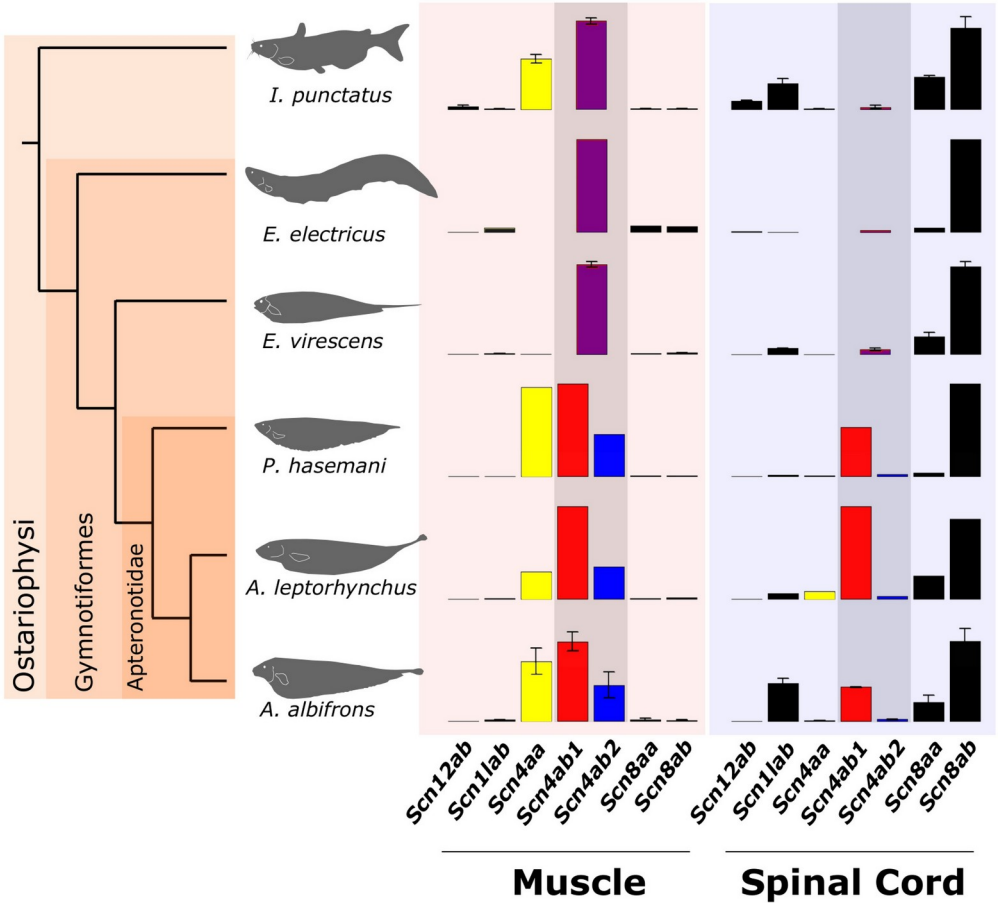
Carlson and Birkett (2022), Fig. 2.18, p. 51.

Electric fish express specialized sodium channels

A study of weakly electric fish examined sodium-channel genes in a lineage whose electric organ develops from specialized spinal neurons. These **electromotor neurons** fire rapidly and contribute to electric discharges. Researchers compared channel gene expression in muscle and spinal cord across electric fish and related species.

A gene normally associated with muscle sodium channels had duplicated in this lineage. One copy acquired expression in the spinal cord. The change placed a channel variant in a new cellular setting, allowing an inherited molecular component to contribute to a specialized neuronal signaling system.

The comparison identified both the gene duplication and a change in where a copy was expressed. These are distinct biological steps. A channel's properties matter only in cells that actually produce it, so changes in expression can be as important as changes in the protein's amino-acid sequence.



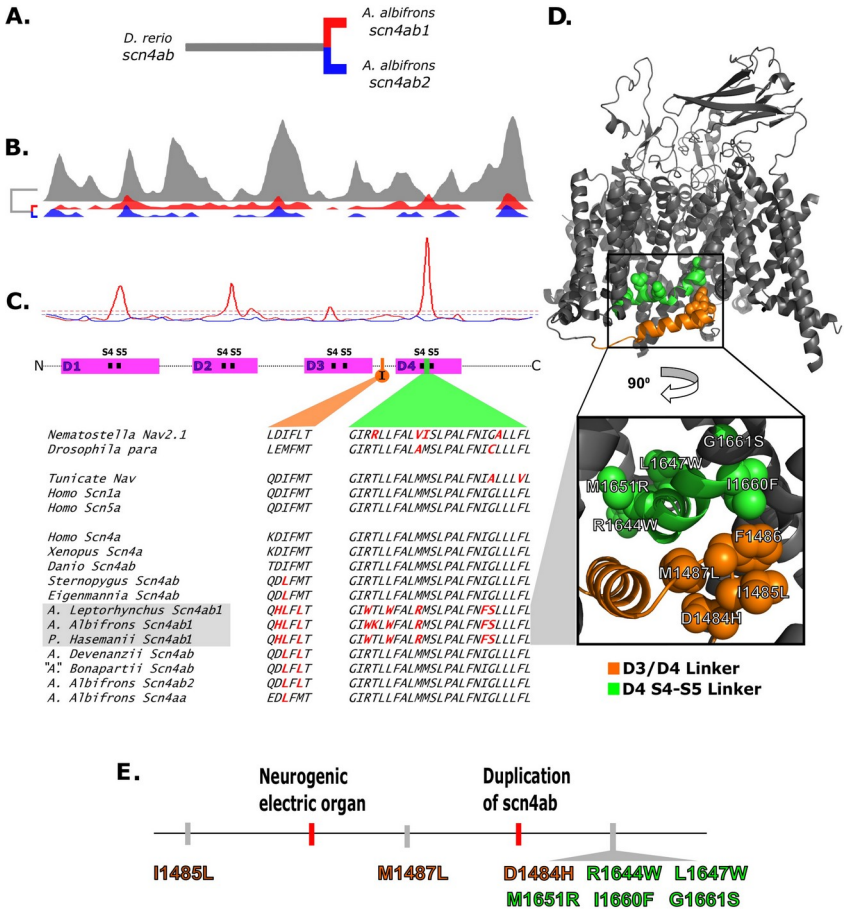
Thompson et al. (2018), Fig. 1. Sodium-channel expression and ionic currents.

Channel inactivation limits a sodium current's duration

The fish study tested how sequence differences affected sodium-channel function. **Inactivation** is a channel state in which conduction stops after activation even though depolarization persists. It helps terminate the inward sodium current and influences how readily a cell can produce another electrical event.

Amino-acid substitutions in the specialized channel produced a persistent sodium current. Some sodium continued to enter during depolarization instead of the current shutting off completely. The experiment identified a specific molecular change that altered the time course of an ionic current through the membrane.

The result explains how related channel proteins can produce different electrical behavior. A channel's selectivity determines which ions cross, while its gating determines when conduction begins and ends. Modifying the inactivation machinery changes the duration of sodium entry without requiring the cell to invent a new kind of ion.



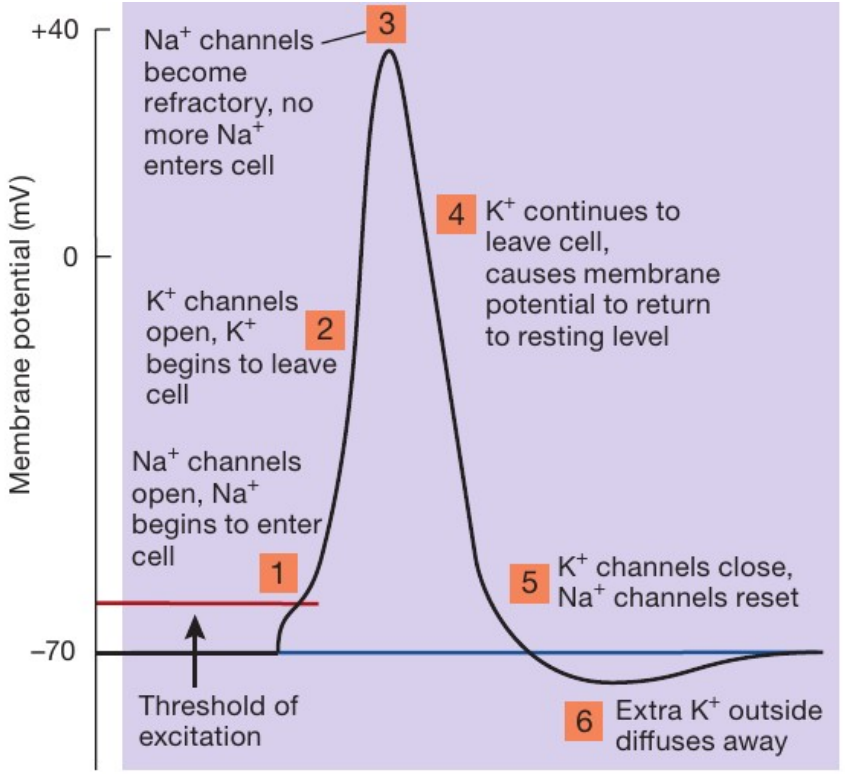
Thompson et al. (2018), Fig. 3. Sodium-channel expression and ionic currents.

Threshold initiates a regenerative action potential

An **action potential** is a rapid, regenerative change in membrane voltage. If depolarization reaches **threshold**, enough voltage-gated sodium channels open to produce further depolarization. That additional change opens still more channels, creating positive feedback during the rising phase of the electrical event.

Sodium enters because the existing concentration and electrical forces favor its inward movement at the start of the event. The membrane becomes less negative and can briefly become positive inside. The rising phase therefore follows the opening of channels that allow an existing ionic gradient to drive current.

An action potential differs from a small local voltage change because it regenerates through channel activation. Once the event begins under normal conditions, increasing the stimulus does not simply make that one spike proportionally taller. Stronger input can instead change when spikes occur and how often the neuron fires.



ɪation can cause them to open again.

lually returns to resting membrane potential as the potassium channels finally

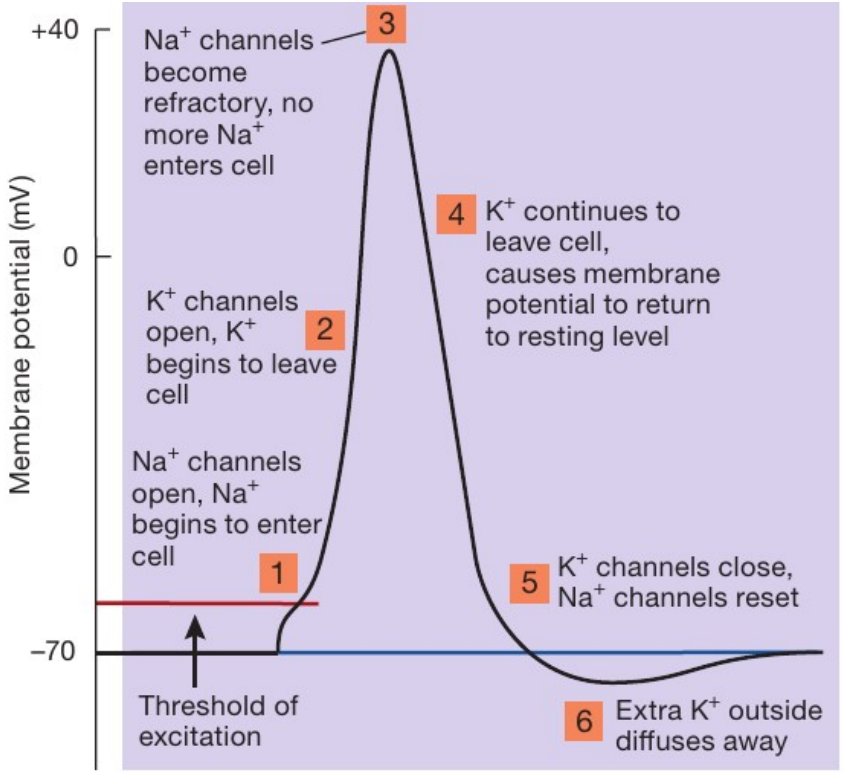
Carlson and Birkett (2022), Fig. 2.20, p. 52.

Sodium and potassium currents restore voltage

During the action potential, sodium channels enter the inactivated state while voltage-gated potassium channels contribute increasing outward current. Potassium leaves the cell, moving positive charge outward. Together, reduced inward sodium current and increased outward potassium current drive the membrane back toward a negative voltage.

Potassium conductance can remain elevated after the spike, briefly making the membrane more negative than its resting value. This **afterhyperpolarization** changes how incoming signals influence the cell immediately afterward. The falling phase and recovery therefore depend on channel kinetics as well as on the original depolarizing stimulus.

Sodium channels must recover from inactivation before they can participate normally in another spike. The resulting **refractory period** limits rapid repeated firing. Electrical recovery involves channel-state changes, while longer-term maintenance of the sodium and potassium concentration gradients depends on the cell's continuing metabolic activity.



ɪation can cause them to open again.

lually returns to resting membrane potential as the potassium channels finally

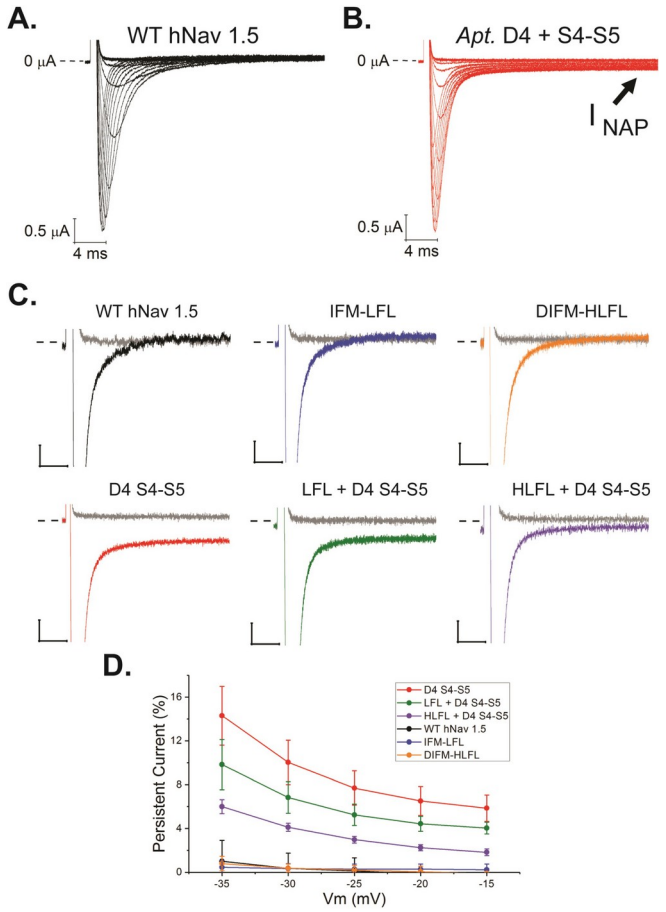
Carlson and Birkett (2022), Fig. 2.20, p. 52.

Persistent sodium current supports extreme firing

Electromotor neurons in the studied fish lineage fire at exceptionally high rates. The investigators tested channel variants carrying substitutions in regions involved in inactivation. The functional measurements tested whether the evolutionary sequence changes altered the current conducted by the protein.

The specialized substitutions produced a persistent component of sodium current. Persistent inward current can help sustain depolarizing drive between electrical events. The authors proposed that this altered gating contributed to the extreme neuronal firing used by the fish's neuron-derived electric organ during electrosensory behavior.

The experiment distinguishes a gene's history, its expression location, and its electrical effect. Each contributes a different part of the explanation. Evolution changed which channel was present in the spinal neurons and how that channel inactivated, providing a molecular route to a specialized neuronal signaling phenotype.



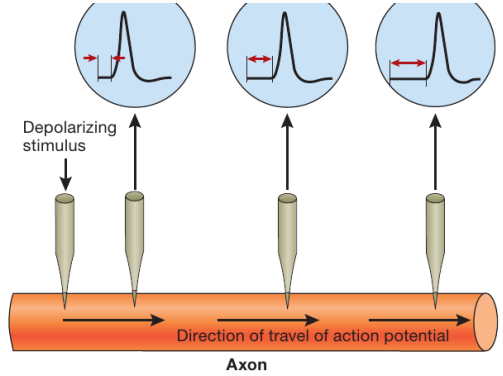
Thompson et al. (2018), Fig. 4. Sodium-channel expression and ionic currents.

An action potential regenerates along an axon

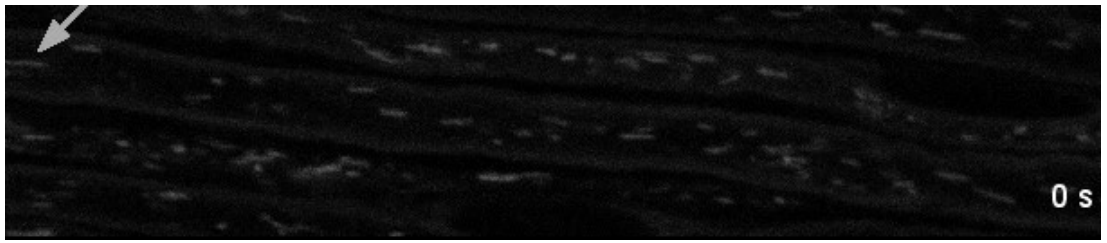
An active patch of axonal membrane produces current that spreads into adjacent regions. If the neighboring membrane reaches threshold, its voltage-gated channels generate another action potential. The signal therefore travels through repeated local regeneration rather than through one electrical event passively fading along the full length of the axon.

The membrane just activated is temporarily refractory, which favors forward propagation under normal conditions. Each newly activated region depolarizes the region ahead. Ion channels distributed along the axon allow the same type of electrical event to be reproduced as it travels away from its starting point.

Conduction and intracellular cargo transport have different mechanisms and speeds. The propagating spike is generated through local membrane currents, while motor proteins move physical materials along internal tracks. A healthy axon needs both: electrical propagation for rapid communication and cargo delivery to maintain the distant membrane and terminal.



Carlson and Birkett (2022), Fig. 2.21, p. 53.



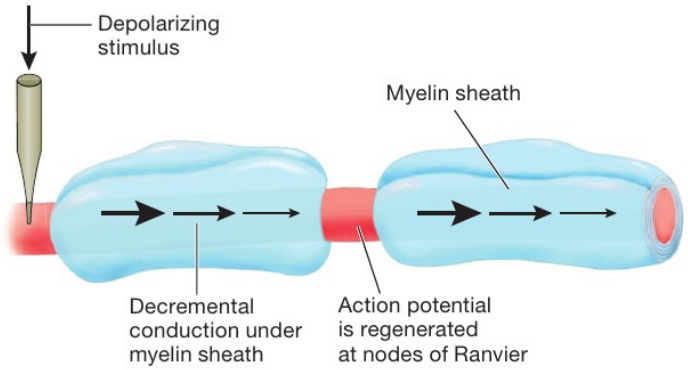
Sajic et al. (2013), pbio.1001754.s004.avi. Aesthetic context: Axonal mitochondrial transport.

Myelin speeds current spread between exposed nodes

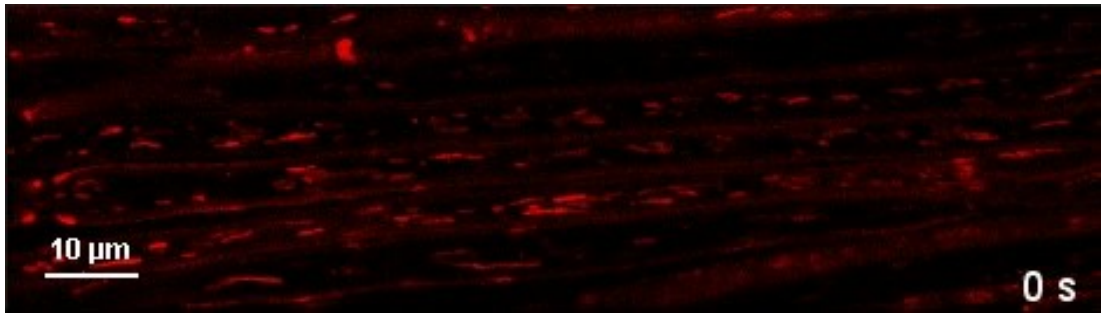
In a myelinated axon, most membrane is covered by insulating glial layers. Current spreads efficiently beneath the sheath and regenerates action potentials at the nodes of Ranvier. This arrangement is called **saltatory conduction** because the active regeneration occurs at separated exposed regions along the axon.

The electrical influence travels through the intervening region, so the signal does not disappear and reappear without a physical route. Myelin reduces current loss and changes the membrane's electrical loading. Channels concentrated at nodes restore the full action potential after the current has spread to the next excitable region.

Loss of myelin can make conduction slower or unreliable. The effect depends on how the exposed membrane and channel distribution change. Maintaining communication therefore requires both appropriate ion-channel function and an intact arrangement of glial insulation and nodal membrane along the same axonal pathway.



Carlson and Birkett (2022), Fig. 2.23, p. 54.



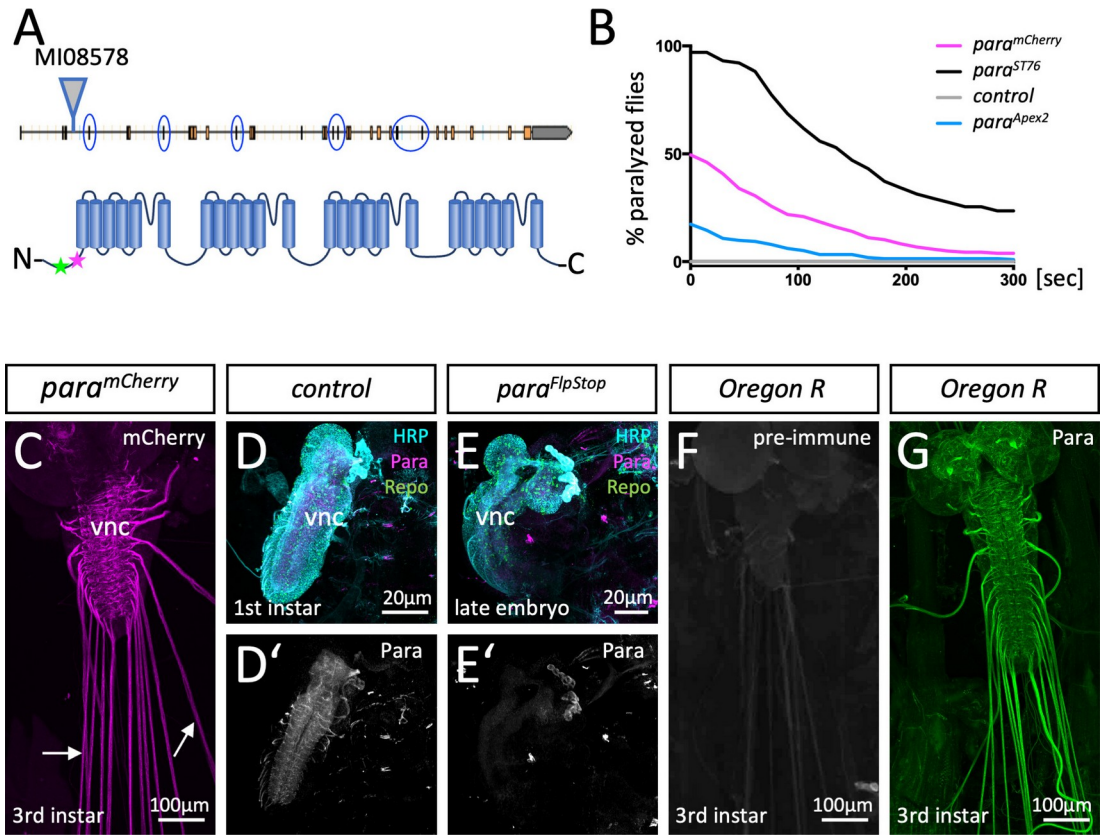
Sajic et al. (2013), pbio.1001754.s005.avi. Aesthetic context: Axonal mitochondrial transport.

Glia organize axonal ion-channel clusters

An experimental study in fruit flies examined how glial cells influence clustering of voltage-gated ion channels along axons. Flies lack vertebrate-style compact myelin. The preparation therefore allowed researchers to investigate glial control of excitable membrane organization separately from the presence of a thick myelin sheath.

The investigators found glial-dependent organization of ion-channel clusters in the fly nervous system. Their results place interactions between neurons and glia within the mechanism that arranges excitable membrane proteins. Supporting cells can influence electrical signaling by affecting where the channels are concentrated along the neuronal surface.

The study broadens the functional comparison between vertebrate myelin and invertebrate glial interactions. Myelin is one specialized way of organizing conduction, while glial regulation of axonal channels is a broader cellular process. The source experiment concerns actual channel distribution, rather than assuming identical axonal architecture across animal groups.



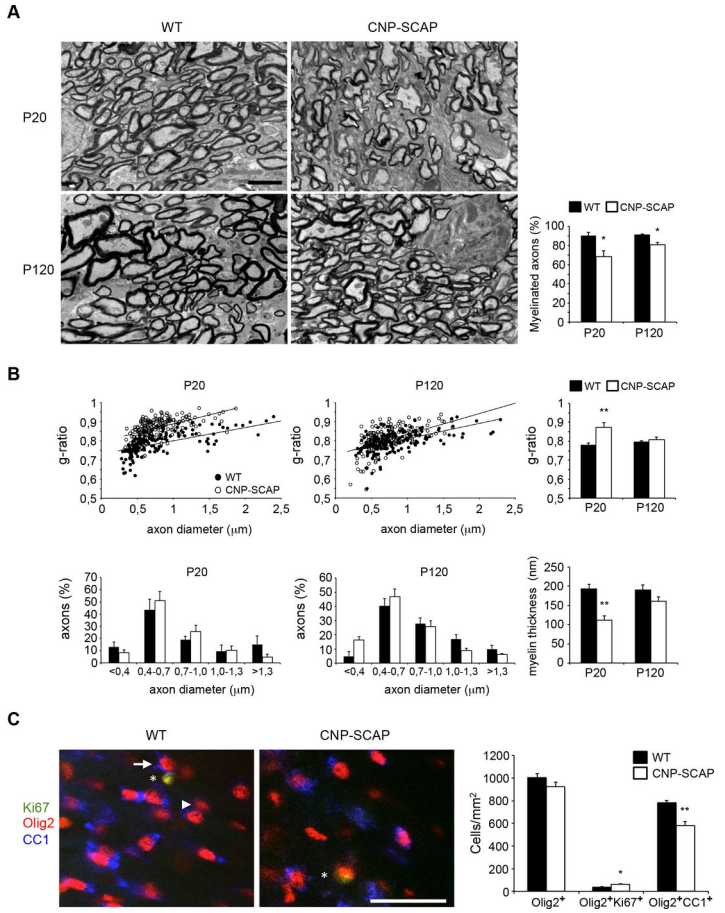
Rey et al. (2023), Fig. 1. Glial interactions and axonal channel organization.

Myelin construction requires several cellular steps

The mouse lipid-synthesis experiments distinguished the contributions of oligodendrocytes from those of astrocytes. Selective genetic manipulations changed which glial population could manufacture particular membrane components. Researchers then measured how myelin developed, testing whether a sheath could be assembled normally under each condition.

Oligodendrocyte lipid synthesis alone did not account for the entire supply used in central myelin. Astrocytic contributions also affected formation of the sheath. The result identifies metabolic exchange between glial populations as one step within the larger process that produces an electrically effective axonal covering.

Building myelin therefore involves more than wrapping an available membrane around an axon. Cells must obtain sufficient lipids, distribute membrane, initiate contacts, and stabilize sheaths. Different experiments isolate different steps, explaining why a defect in metabolism can alter conduction even when the axon's own channel proteins remain present.



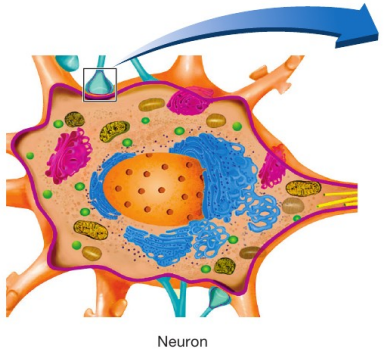
Camargo et al. (2017), Fig. 2. Glial lipid supply and central myelin.

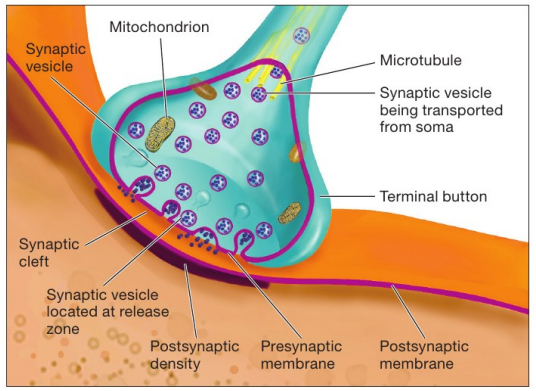
Synapses separate the sending and receiving membranes

A **synapse** is a specialized junction through which one cell influences another. At a chemical synapse, the **presynaptic** cell releases a transmitter and the **postsynaptic** cell carries receptors that respond to it. The two membranes remain separated by a narrow extracellular space called the **synaptic cleft**.

An arriving action potential depolarizes the presynaptic terminal. The resulting sequence causes transmitter-containing vesicles to release their contents into the cleft. Transmitter molecules diffuse across the short distance and bind to receptors, changing the receiving cell's conductance or activating signaling processes within that cell.

The structure provides distinct locations at which communication can change. A terminal can alter transmitter release, the cleft can change how rapidly transmitter is removed, and the receiving membrane can alter receptor number or function. A different postsynaptic response can therefore arise from changes on either side of the junction.





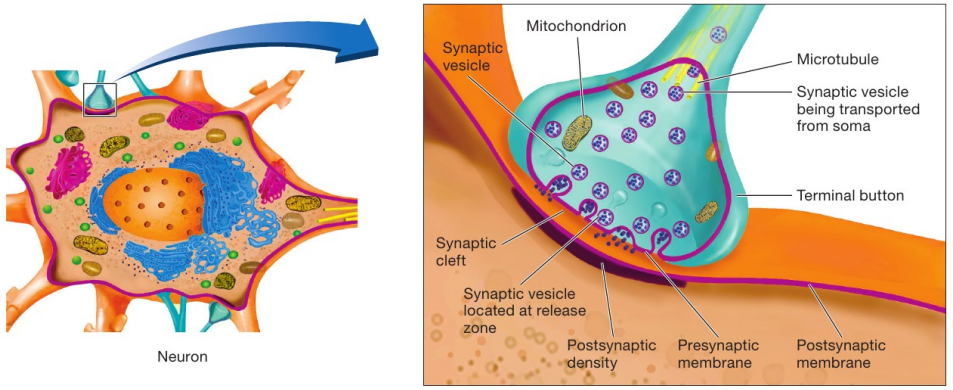
Carlson and Birkett (2022), Fig. 2.26, p. 57.

Synaptic vesicles package chemical messengers

Synaptic vesicles are small membrane compartments containing neurotransmitter. They collect near specialized release regions of the terminal. The vesicle membrane can fuse with the terminal's membrane, opening a route through which the packaged molecules enter the extracellular synaptic cleft and reach the receiving cell.

A terminal maintains vesicles in different functional states rather than releasing its entire supply with every spike. Some are positioned and prepared for rapid fusion, while others replenish the available population. Maintaining these supplies requires local membrane handling and the delivery of vesicle components from elsewhere in the neuron.

Transported vesicle precursors and release-ready synaptic vesicles are related but distinct stages of supply. Materials travel through the axon before becoming part of the terminal's working machinery. The availability, positioning, and readiness of vesicles all influence how a presynaptic electrical event becomes a chemical signal.



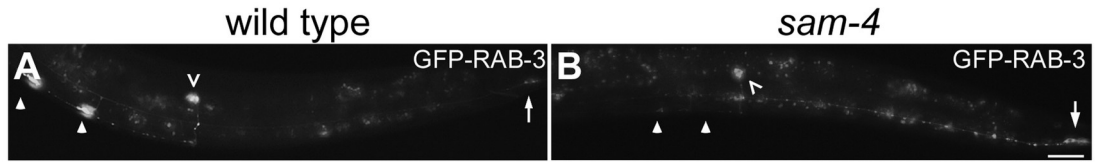
Carlson and Birkett (2022), Fig. 2.26, p. 57.

A vesicle protein controls how far cargo travels

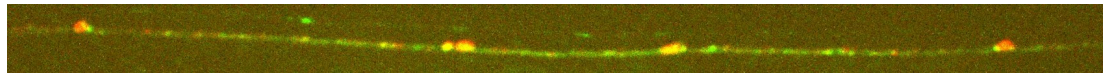
A study in the nematode **Caenorhabditis elegans** identified the vesicle-associated protein **SAM-4** as a regulator of axonal transport. Researchers labeled vesicle components and followed their movement along neurons. Genetic manipulations allowed them to compare cargo movement when SAM-4 function was present or disrupted.

Loss of SAM-4 reduced transport **processivity**, meaning that cargo traveled shorter distances before a run ended. Its movement speed during a run was not reduced in the same way. Distance per run and speed are therefore distinct properties of the motor-driven delivery system.

The investigators examined interactions with **UNC-104**, the worm version of a kinesin motor involved in synaptic vesicle transport. Motor mutations that increased processivity suppressed the transport defect. The experiment identifies the regulation of sustained cargo movement as a mechanism contributing to the supply of presynaptic components.



Zheng et al. (2014), Fig. 1A–B. SAM-4 and synaptic vesicle transport.



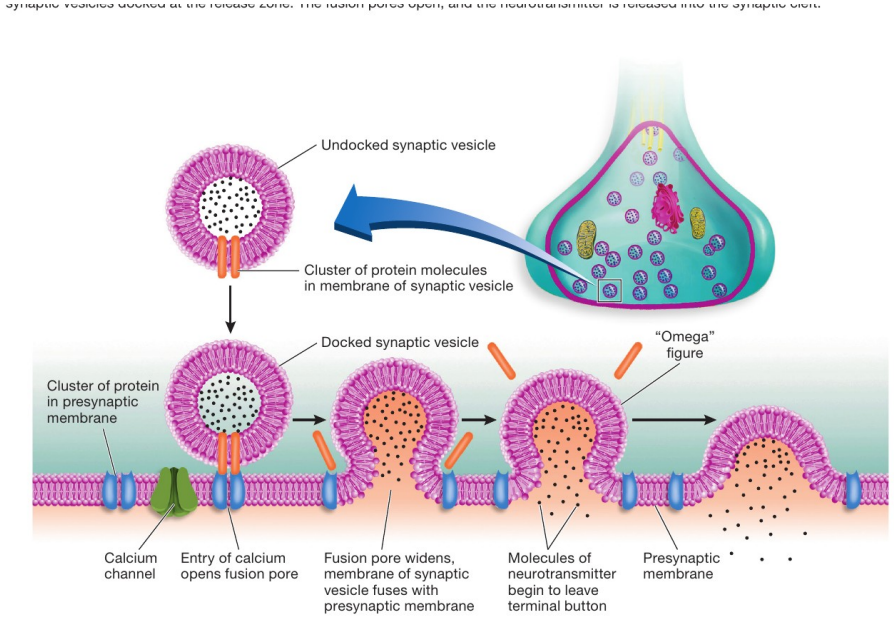
Zheng et al. (2014), pgen.1004644.s014.avi. Academic: Two-color vesicle transport.

Presynaptic calcium triggers vesicle fusion

When an action potential reaches the terminal, depolarization opens voltage-gated calcium channels. Calcium enters because its concentration is much higher outside the cell and the electrical forces favor inward movement. The local increase near release machinery triggers the fusion of prepared synaptic vesicles with the presynaptic membrane.

Vesicle fusion releases transmitter into the synaptic cleft through **exocytosis**. The arriving spike and the release event are therefore different physical processes. A voltage change opens channels, calcium enters, fusion machinery responds, and transmitter then reaches receptors on the postsynaptic cell.

The exact arrangement matters because calcium acts locally near its entry sites. A vesicle close to a calcium channel can encounter a different calcium signal from a more distant vesicle. The number of prepared vesicles and their positions relative to channels can both alter the response to the same arriving spike.



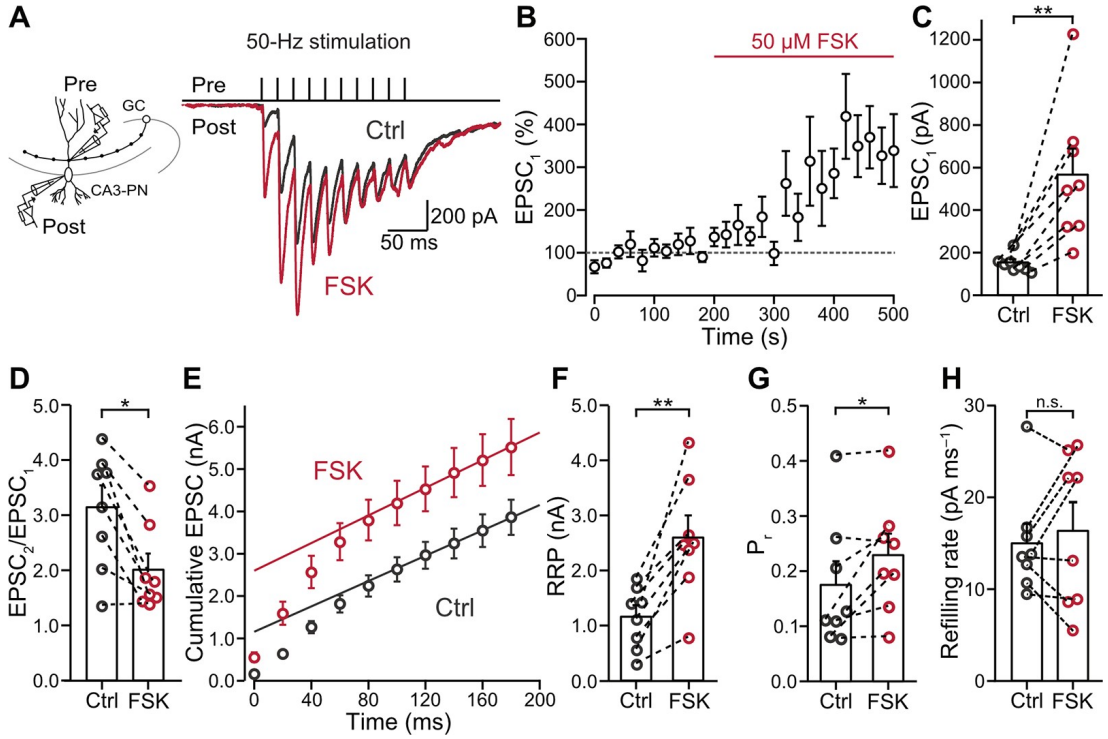
Carlson and Birkett (2022), Fig. 2.27, p. 58.

Potentiation enlarges the pool of release-ready vesicles

A 2024 study examined the **mossy fiber synapse**, where hippocampal granule-cell axons contact CA3 neurons. Researchers recorded from the presynaptic terminal and postsynaptic neuron while inducing chemical potentiation. They used this preparation to ask which presynaptic changes could strengthen synaptic transmission.

The treatment increased the **readily releasable pool**, the set of vesicles prepared to fuse rapidly. It also increased vesicular release probability. A stronger postsynaptic response therefore reflected changes in the supply and behavior of presynaptic vesicles rather than simply a larger incoming electrical spike.

A vesicle pool is a functional category, not a separate organelle. Its size depends on how many vesicles are prepared to participate in rapid release under the tested conditions. Measuring that pool helps separate the number of available release opportunities from the likelihood that each opportunity produces fusion.



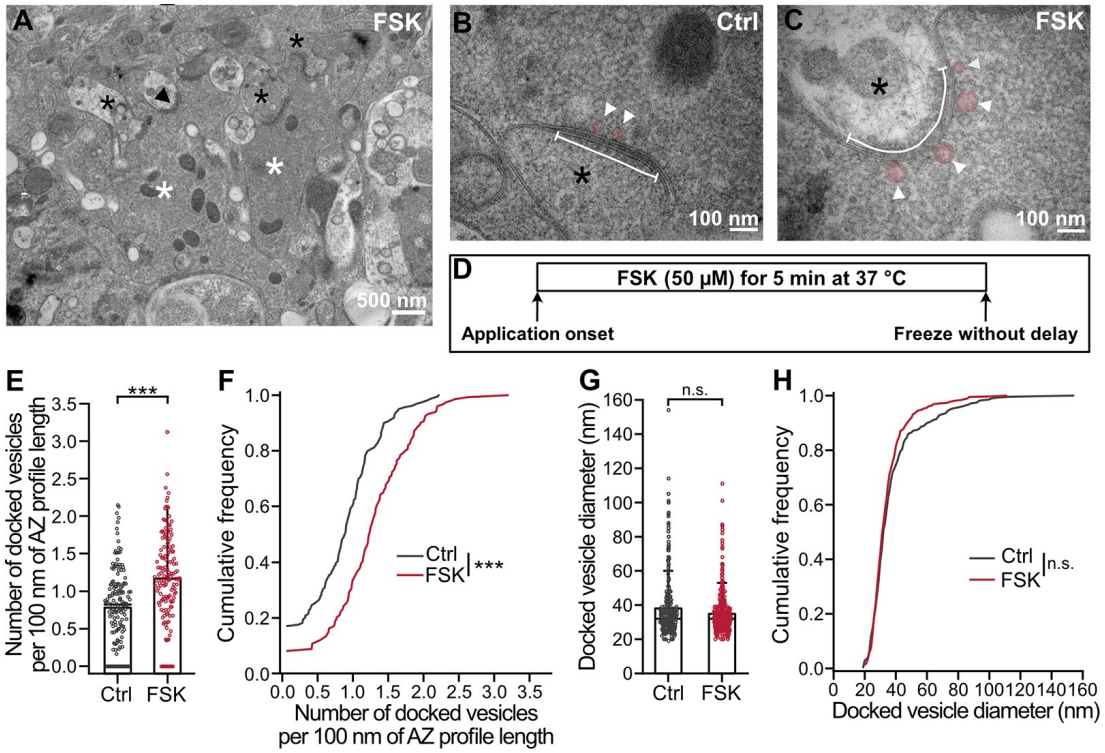
Kim et al. (2024), Fig. 1. Presynaptic vesicles and release-site organization.

Potentiated terminals contain more docked vesicles

The investigators combined electrical recordings with electron microscopy of hippocampal terminals. Electron microscopy resolves structures too small to distinguish with ordinary light microscopy. They examined how many vesicles were close to the presynaptic membrane after the same treatment that strengthened synaptic transmission.

Potentiated terminals contained more vesicles near the membrane and more docked vesicles. The anatomical measurements therefore identified a structural change corresponding to the increased release-ready pool measured physiologically. More vesicles were positioned where the fusion machinery could prepare them for rapid transmitter release.

Docking and release remain distinct events. A membrane-adjacent vesicle still requires the molecular conditions that permit fusion. Combining structural and physiological measurements helps explain how a terminal becomes more effective: it can position more vesicles at release sites and change the probability that prepared vesicles are released.



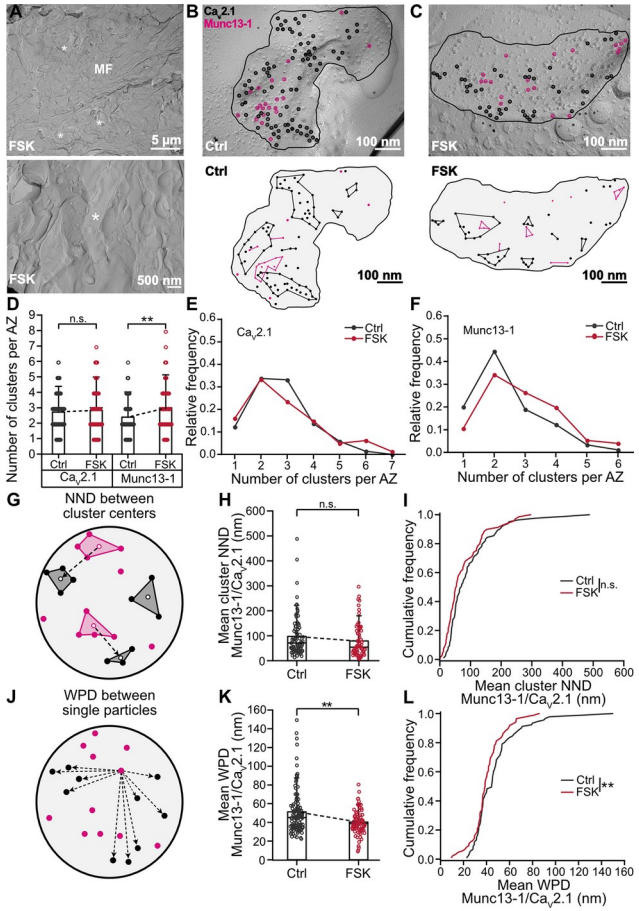
Kim et al. (2024), Fig. 3. Presynaptic vesicles and release-site organization.

Calcium channels approach vesicle-priming sites

The study used freeze-fracture labeling to examine proteins within presynaptic release regions. Researchers measured the distribution of **CaV2.1** calcium channels and **Munc13-1**, a protein involved in preparing vesicles for release. The method addressed the arrangement of molecules within the terminal membrane.

Chemical potentiation reduced the distance between channel clusters and priming-protein clusters. The terminal therefore changed not only the number of prepared vesicles but also the organization of release machinery relative to calcium entry. Small spatial changes occurred within the region that converts voltage into transmitter release.

The authors proposed that this reorganization improves coupling between calcium entry and vesicle fusion. The proposal concerns the local architecture of the release site. It provides a physical mechanism through which a synapse can respond more strongly to an electrical event without changing the identity of its neurotransmitter.



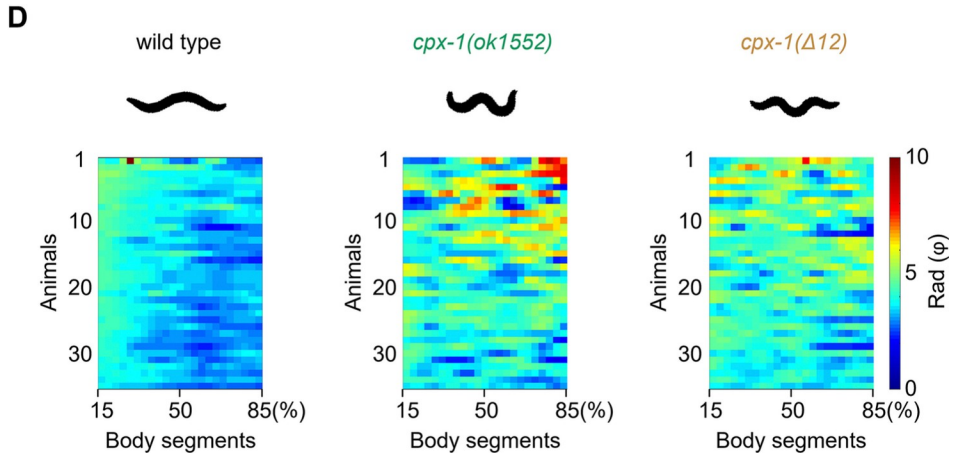
Kim et al. (2024), Fig. 4. Presynaptic vesicles and release-site organization.

Complexin regulates spontaneous and evoked release

A 2025 study examined **complexin**, a protein that regulates synaptic vesicle fusion, at worm neuromuscular junctions. These junctions transmit signals from motor neurons to muscle. The investigators combined recordings of synaptic currents with genetic changes and measurements of the animals' body curvature and movement.

Loss of complexin increased spontaneous release at excitatory synapses and altered evoked transmission. **Spontaneous release** occurs without the experimenter triggering a presynaptic event, while **evoked release** follows an applied stimulus. The same protein therefore affected different aspects of the terminal's release behavior.

Mutant worms also moved abnormally, allowing the cellular measurements to be considered alongside motor output. The study investigated how changes in release regulation disturb the balance of excitation and inhibition reaching muscles. It did not treat all transmitter release as one process with a single unchanged rate.



Wang et al. (2025), Fig. 1D. Release regulation and worm neuromuscular function.



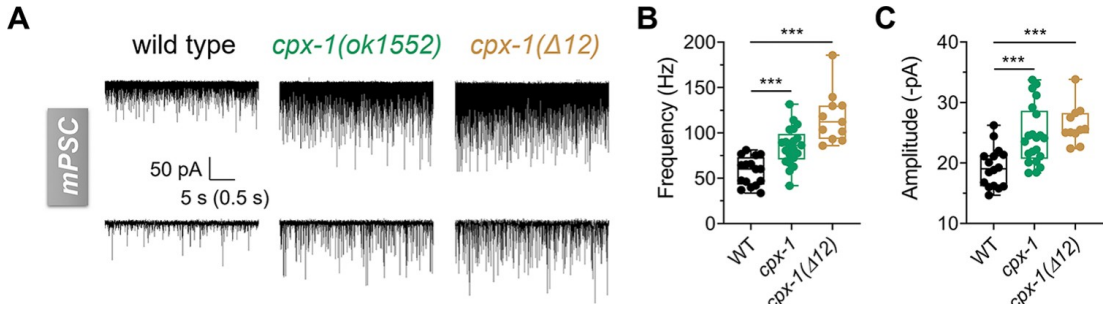
Wang et al. (2025), pbio.3003023.s009.avi. Aesthetic context: Locomotion and body curvature.

Different complexin mutations separate release functions

The investigators compared a complexin-null mutation with a mutation designed to impair its clamping function while preserving evoked release. A **clamp** limits fusion under particular conditions. Comparing these manipulations separated a protein's role in suppressing unwanted release from its role during a triggered response.

The clamping mutation increased spontaneous excitatory release while retaining evoked responses. The null mutation produced a different combination of defects. Removing an entire protein therefore had a broader effect than changing one part of its function, even though both manipulations affected the same release regulator.

The distinction connects molecular organization to behavior through a specific chain of events. A protein domain changes fusion control, the pattern of transmitter release changes, and muscles receive altered input. Comparing mutations helps identify which part of that chain produces the observed disturbance in coordinated movement.



Wang et al. (2025), Fig. 2A–C. Release regulation and worm neuromuscular function.



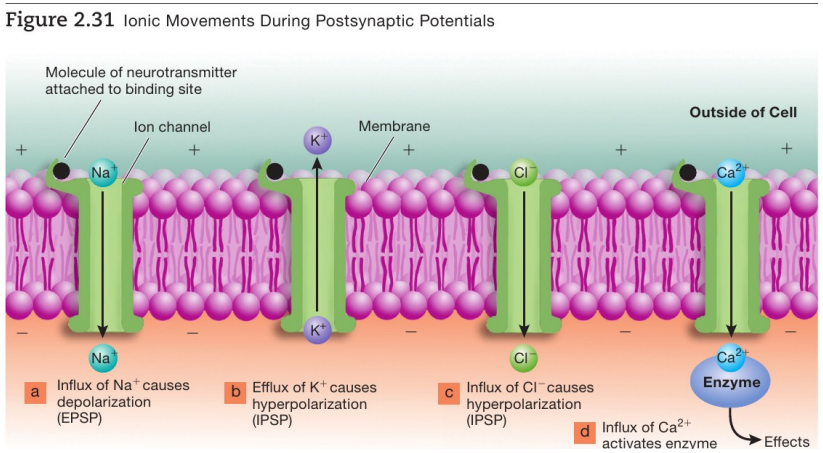
Wang et al. (2025), pbio.3003023.s010.avi. Academic: Locomotion and body curvature.

Excitatory input moves a cell toward firing threshold

An **excitatory postsynaptic potential**, or **EPSP**, is a local voltage change that increases the likelihood of an action potential. Transmitter binding opens receptor-associated pathways that alter membrane conductance. The effect depends on the ions that can cross and the membrane voltage at that time.

An EPSP is usually smaller than an action potential and varies with input strength. It can spread away from its point of origin and become weaker with distance. Several EPSPs occurring close together can combine, changing the voltage reaching regions that initiate the neuron's outgoing electrical signal.

Excitation is a functional description, not a claim that every receptor for a particular transmitter always has the same effect. Receptor type and ionic gradients determine the response. At a neuromuscular junction, excitatory motor-neuron input can depolarize muscle and contribute to the activation of its contractile machinery.



Carlson and Birkett (2022), Fig. 2.31, p. 61.



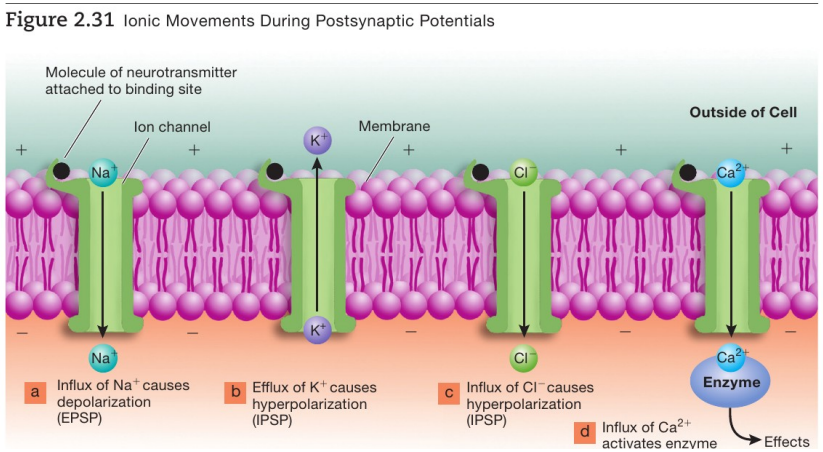
Wang et al. (2025), pbio.3003023.s011.avi. Academic: Locomotion and body curvature.

Inhibition reduces the effect of excitatory input

An **inhibitory postsynaptic potential**, or **IPSP**, makes a neuron less likely to reach firing threshold. Inhibitory receptors can increase conductance to ions such as chloride or potassium. The resulting effect depends on their ionic gradients and the receiving cell's existing membrane voltage.

Inhibition need not always produce a large negative deflection. Increasing conductance can reduce the effect of a simultaneous excitatory current, making depolarization less effective. The electrical influence of an inhibitory synapse therefore depends on its timing, location, and the state of the surrounding membrane.

Muscle coordination also requires appropriately timed inhibitory signaling in many circuits. Excitation can activate one part of the motor apparatus while inhibition helps regulate competing activity. The balance is produced through specific synapses and receptors, rather than by one general chemical signal that simply switches the whole nervous system off.



Carlson and Birkett (2022), Fig. 2.31, p. 61.



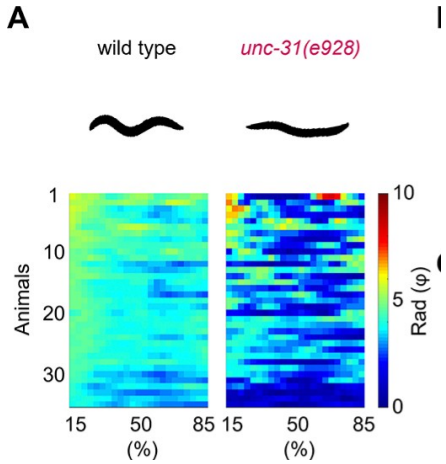
Wang et al. (2025), pbio.3003023.s014.avi. Aesthetic context: Locomotion and body curvature.

CAPS affects excitation and inhibition unequally

The worm study examined **CAPS**, encoded by **unc-31**, another conserved regulator of secretion. Researchers compared excitatory cholinergic transmission with inhibitory GABAergic transmission at neuromuscular junctions. **GABA** is a transmitter that mediates inhibition in this preparation, while acetylcholine drives excitation at the studied excitatory contacts.

Removing UNC-31 reduced evoked excitatory transmission more strongly than evoked inhibitory transmission. Spontaneous release was comparatively preserved under the tested conditions. The mutation therefore altered the balance between two inputs rather than reducing every type of synaptic communication by an identical amount.

The animals' movement depended on that balance. A molecular regulator acting at release machinery could change how opposing motor inputs reached muscle. Measuring both transmitter systems was necessary to explain the behavioral phenotype, because an excitatory recording alone would omit the differently affected inhibitory contribution.



Wang et al. (2025), Fig. 3A. Release regulation and worm neuromuscular function.



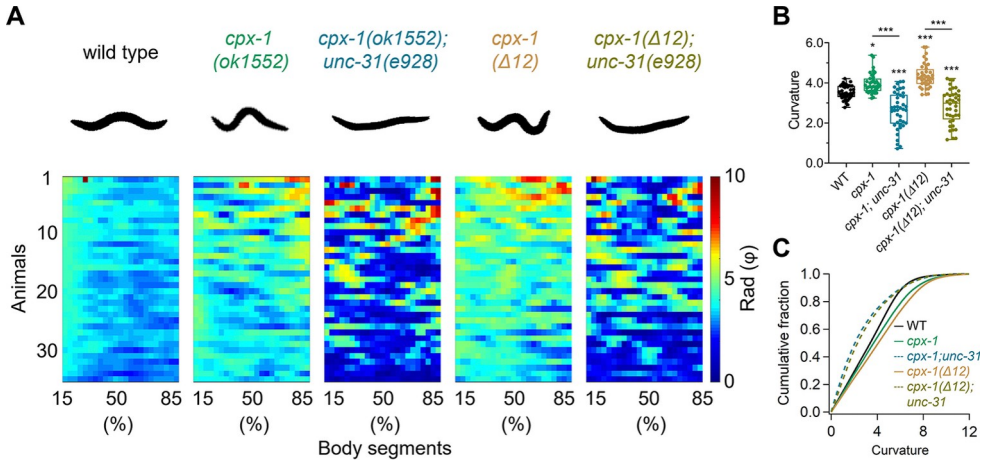
Wang et al. (2025), pbio.3003023.s012.avi. Academic: Locomotion and body curvature.

Genetic interactions change release and movement

The investigators combined complexin and UNC-31 mutations to test whether the proteins' effects were independent. They measured synaptic currents and movement in single and double mutants. This approach examined how one molecular manipulation changed the outcome produced by another within the same release system.

The increase in spontaneous excitatory release caused by complexin loss depended on UNC-31. Double-mutant results therefore differed from the simple addition of two isolated defects. Release regulators acted within an interacting cellular system, making the effect of one protein dependent on the state of another pathway.

The study also tested calcium dependence, connecting the genetic interactions to the ionic trigger for secretion. The behavioral output emerged from these combined changes in release regulation and excitatory–inhibitory balance. Molecular interactions at terminals thus produced consequences that could be measured in the movement of the whole animal.



Wang et al. (2025), Fig. 4A–C. Release regulation and worm neuromuscular function.



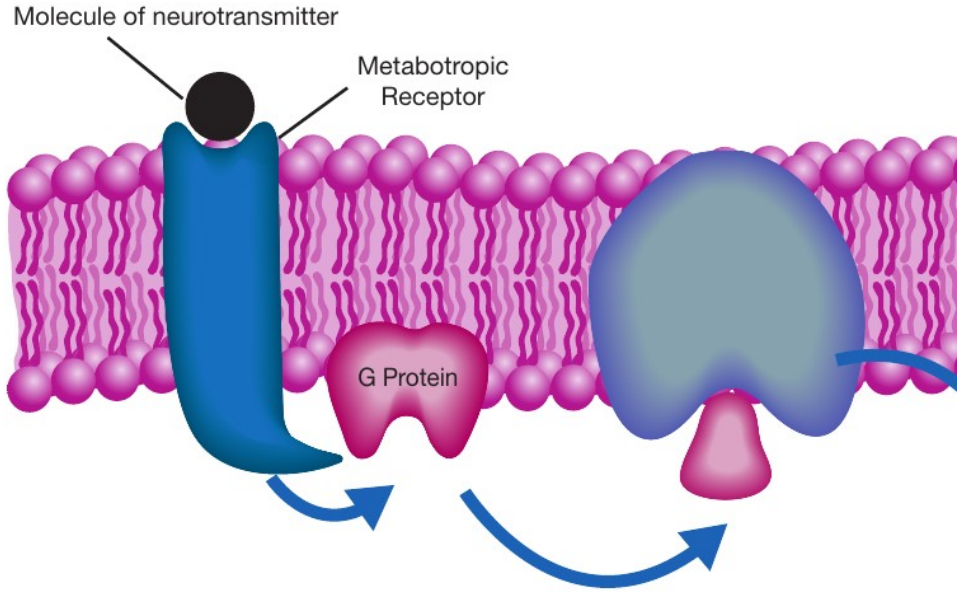
Wang et al. (2025), pbio.3003023.s013.avi. Academic: Locomotion and body curvature.

Receptors convert transmitter binding into a response

An **ionotropic receptor** includes an ion channel that changes state when transmitter binds. The resulting ionic current can rapidly alter membrane voltage. A **metabotropic receptor** instead activates an intracellular signaling pathway, often through a G protein, which can affect channels or other cellular processes indirectly.

Both receptor classes begin with a chemical messenger binding to a specific protein. Their downstream mechanisms differ, so their response timing and duration can differ too. The same transmitter can therefore produce different cellular effects when it acts on receptors with different structures and signaling partners.

Receptors do not respond solely because transmitter is nearby. The messenger must bind to the appropriate molecular site and alter receptor function. The receiving cell's existing channels, internal signaling machinery, and ionic gradients then determine the physical response produced by that binding event.



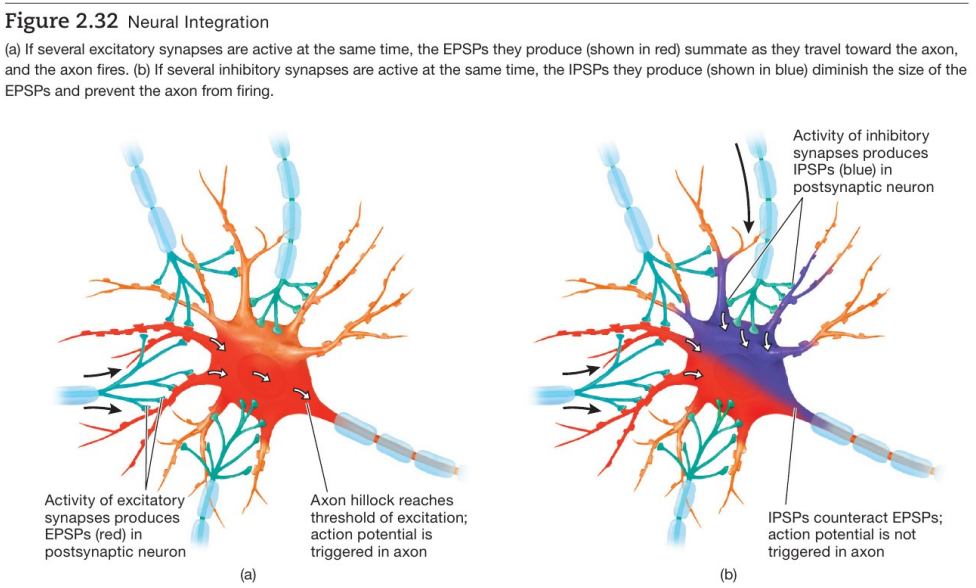
Carlson and Birkett (2022), Fig. 2.30, p. 60.

Synaptic inputs combine across space and time

Spatial summation occurs when inputs arriving at different membrane locations combine. **Temporal summation** occurs when inputs arrive closely enough in time that one voltage change has not ended before another begins. Both processes affect the membrane voltage produced by a sequence of synaptic events.

The influence of a synapse depends on more than its transmitter alone. Its distance from the spike-generating region, the electrical properties of intervening dendrites, and nearby inhibitory input all matter. Two equal currents at different locations can therefore have different effects on the neuron's outgoing signal.

Neural integration is the process through which these inputs jointly affect activity. A neuron does not simply count the number of active synapses. It combines currents in a membrane containing selective conductances, making the timing and arrangement of those inputs part of the computation.



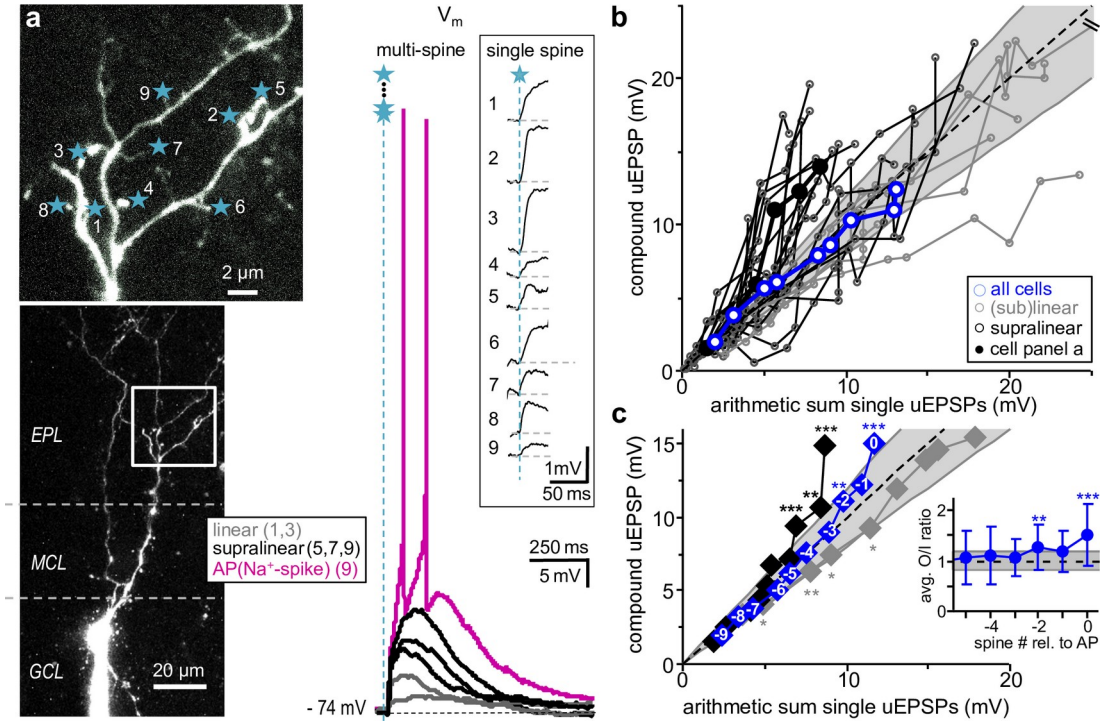
Carlson and Birkett (2022), Fig. 2.32, p. 62.

Several active spines can recruit dendritic spikes

A study of rat olfactory-bulb granule cells activated selected dendritic spines using patterned light stimulation of a caged transmitter compound. **Uncaging** releases an active chemical at a controlled location. The preparation allowed researchers to compare responses as increasing numbers of spines were activated together.

Small groups of active spines produced local and regional signals, while larger groups could recruit a dendritic spike. The investigators measured both the somatic membrane potential and calcium responses. Dendritic activity therefore changed qualitatively as the number of coactive inputs increased within the same neuron.

The result separates passive addition from active membrane recruitment. A group of inputs can depolarize the dendrite enough to engage voltage-dependent mechanisms, producing a response larger than expected from simply adding isolated inputs. The dendrite participates in signal processing rather than acting only as a receiving cable.



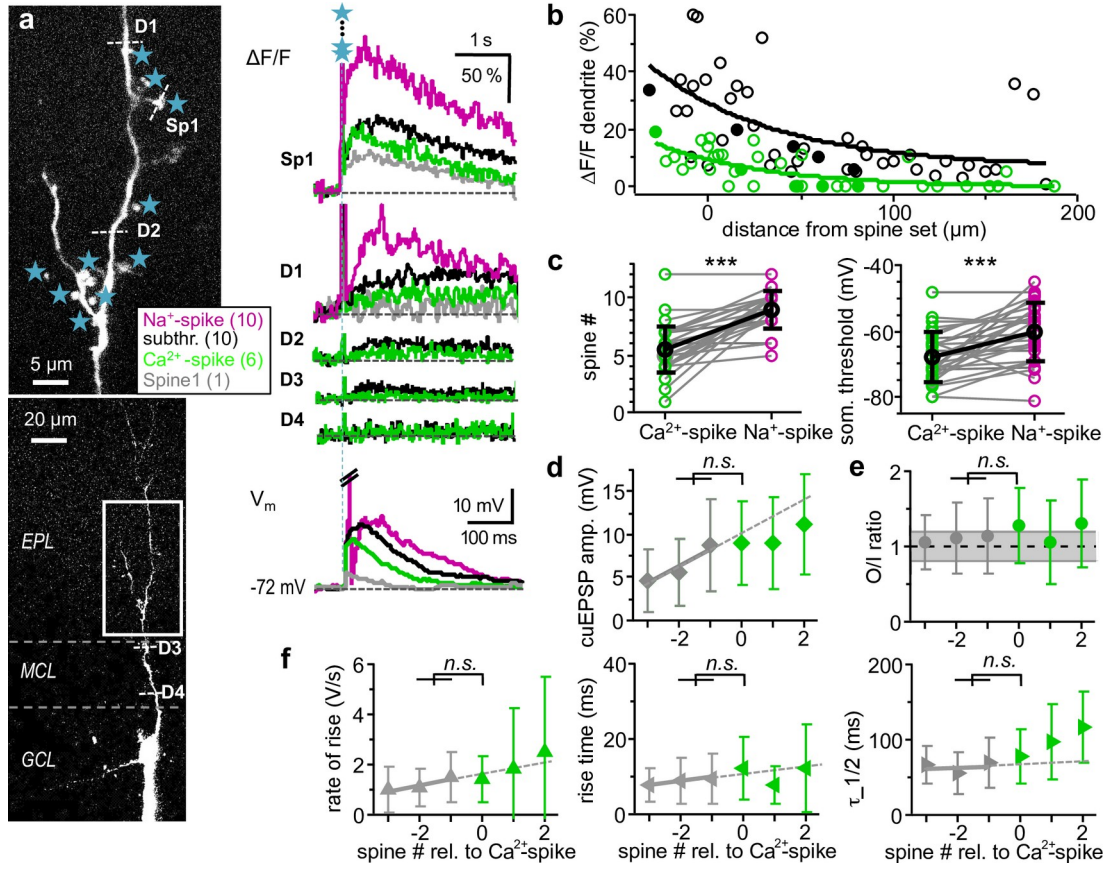
Mueller et al. (2020), Fig. 1. Spine activation and dendritic responses.

Dendritic responses can become larger than linear sums

The researchers compared combined spine activation with the responses expected from individual inputs. A **linear sum** is the result of adding the separate responses directly. A **supralinear** response exceeds that prediction because the combined input engages additional mechanisms within the membrane.

Integration was mostly linear until a threshold number of activated spines recruited stronger dendritic events. Some cells also produced broader sodium-dependent spikes. The same collection of synapses could therefore operate differently depending on how many inputs arrived together and whether an active threshold was crossed.

These measurements explain why timing matters for neuronal communication. Inputs that overlap can recruit conductances that isolated inputs fail to activate. The resulting output is shaped by both the synaptic current entering the cell and the voltage-dependent channels engaged by the combined depolarization.



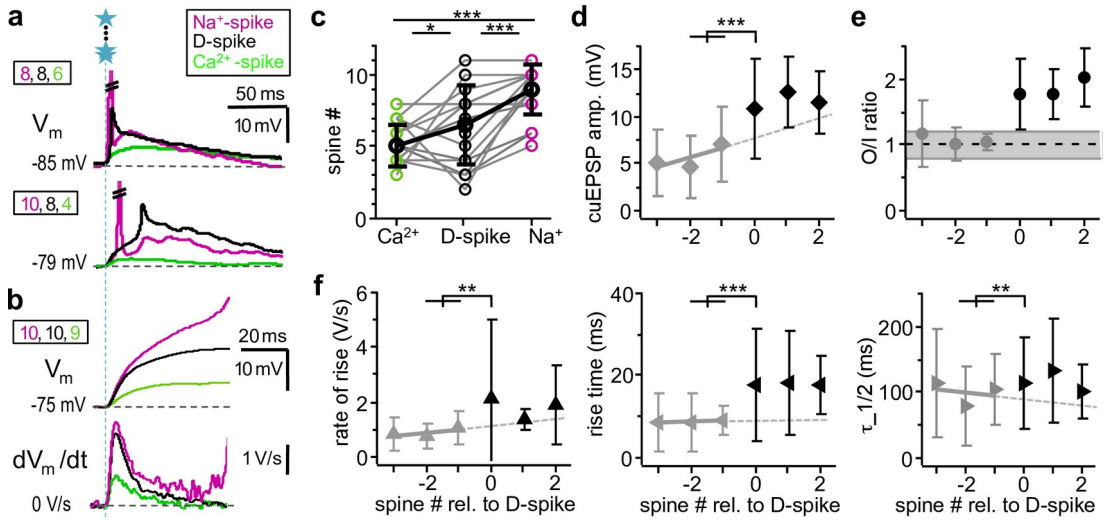
Mueller et al. (2020), Fig. 2. Spine activation and dendritic responses.

NMDA receptors contribute to active dendritic integration

The investigators used receptor and channel manipulations to identify mechanisms underlying supralinear dendritic responses. **NMDA receptors** are glutamate receptors whose activity depends on both transmitter binding and membrane conditions. They can admit calcium and contribute to excitation when the appropriate conditions are met.

NMDA receptors strongly contributed to the nonlinear responses in the studied granule cells. Dendritic voltage-gated sodium and calcium channels also participated. The response therefore emerged from several cooperating membrane mechanisms rather than from one transmitter molecule simply producing the same fixed voltage change at every input strength.

The experiments distinguished receptor involvement from the anatomical appearance of a spine. A spine provides the contact site, but its electrical effect depends on the proteins present in its membrane and nearby dendrite. Cellular structure and channel function jointly determine how the incoming signal influences the rest of the neuron.



Mueller et al. (2020), Fig. 3. Spine activation and dendritic responses.

Reuptake and enzymes terminate transmitter action

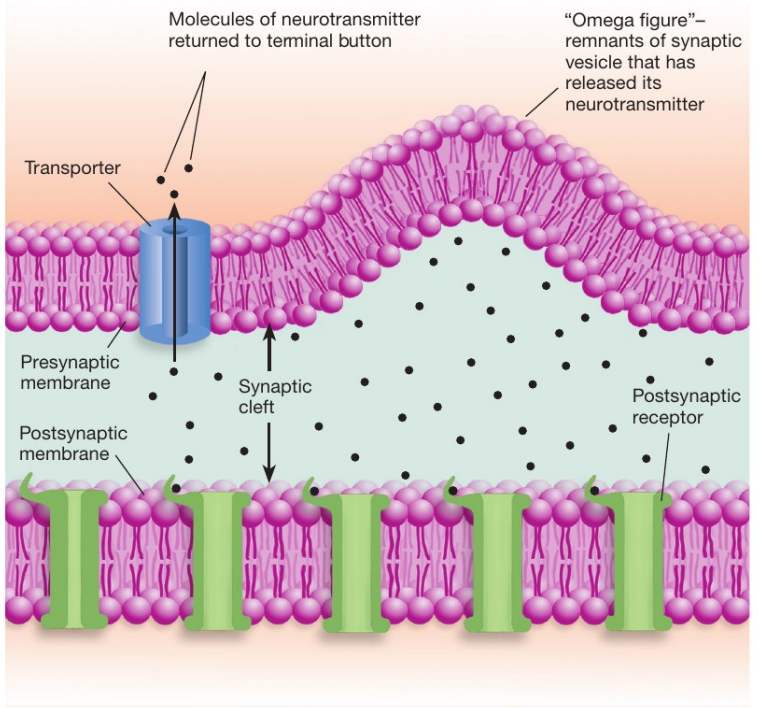
A chemical signal must end as well as begin. **Reuptake** returns released transmitter to cells through transport proteins, reducing its concentration outside. Other transmitters are broken down by enzymes. These mechanisms limit how long receptors remain exposed to released messenger in and around the synapse.

At cholinergic synapses, **acetylcholinesterase** breaks down acetylcholine. Inhibiting this enzyme can prolong transmitter action because molecules remain available to activate receptors. The effect follows altered chemical clearance, even when the presynaptic electrical event and the initial amount released are unchanged.

Myasthenia gravis is an autoimmune disease that can impair communication at neuromuscular junctions. Reducing acetylcholine breakdown prolongs its availability to receptors and can improve muscle activation. The treatment changes signal clearance rather than repairing the underlying immune attack, separating a temporary physiological improvement from removal of the disease mechanism.

Figure 2.33 Reuptake

Molecules of a neurotransmitter released into the synaptic cleft are transported back into the presynaptic terminal button.



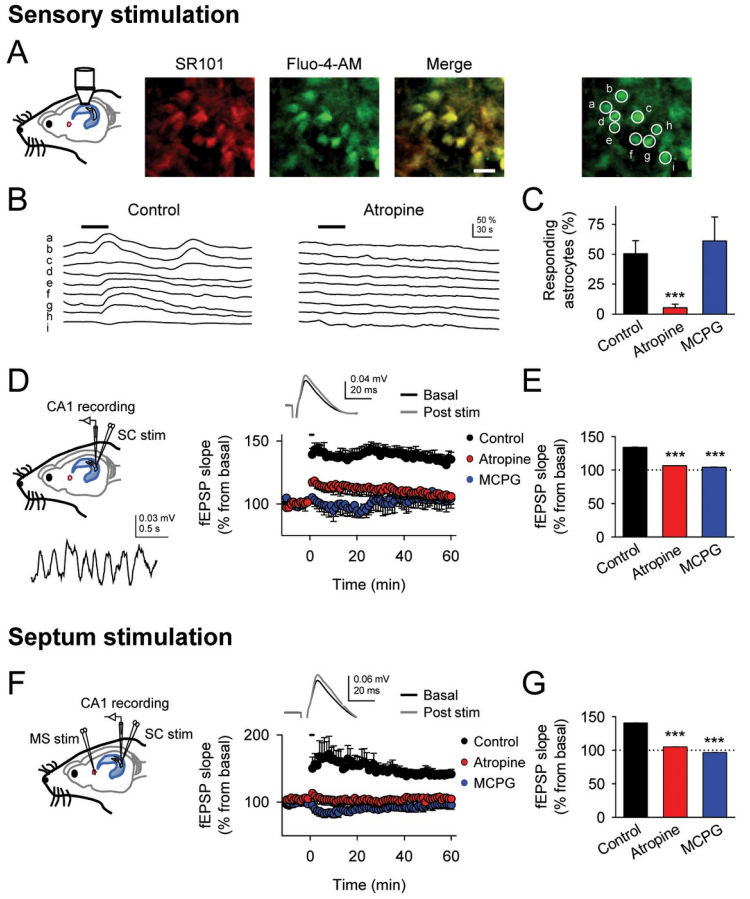
Carlson and Birkett (2022), Fig. 2.33, p. 63.

Astrocytes participate in cholinergic synaptic plasticity

In mouse hippocampal experiments, cholinergic stimulation raised calcium in astrocytes while strengthening transmission between CA3 and CA1 neurons. The investigators combined stimulation, receptor blockers, and cellular recordings. These methods tested how acetylcholine recruited both neuronal and glial responses within one synaptic pathway.

Muscarinic acetylcholine receptors and metabotropic glutamate receptors participated in the potentiation. The receptor classes activate intracellular pathways rather than simply opening one rapid transmitter-gated pore. The experiment therefore places slower receptor signaling alongside the fast electrical events that transmit individual spikes between neurons.

The result identifies glia as active participants in a particular form of plasticity. The astrocyte's calcium signal contributed to a persistent change in synaptic efficacy. The pathway's behavior depended on signaling across several cell types and receptor systems, rather than only on the presynaptic axon and postsynaptic membrane.



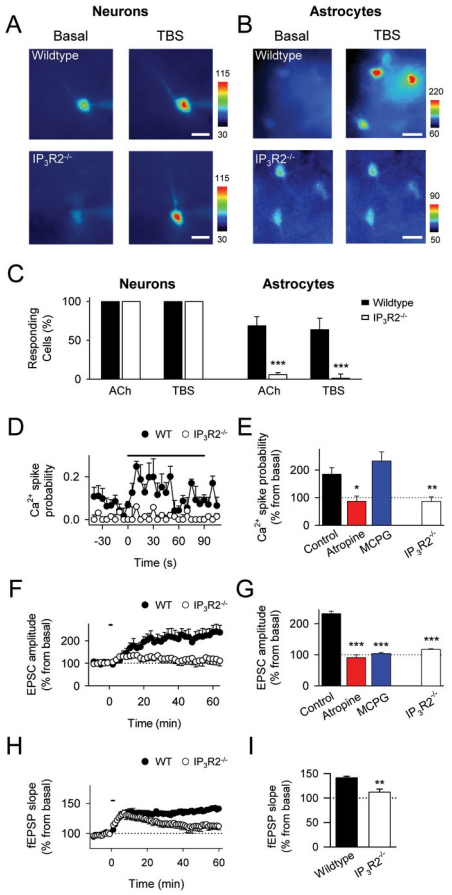
Navarrete et al. (2012), Fig. 1. Astrocytic calcium and hippocampal potentiation.

Coincident glial and neuronal activity strengthens input

Navarrete and colleagues suppressed astrocytic calcium signaling and tested the effect on cholinergically induced potentiation. They then combined a restored astrocytic calcium rise with postsynaptic depolarization. The design separated the contribution of the glial signal from the electrical state of the receiving neuron.

Suppressing the astrocytic signal prevented potentiation, while the paired restoration recovered it under the experimental conditions. Coincidence between the cellular events therefore influenced whether the synapse changed persistently. A presynaptic pathway could produce different lasting effects depending on concurrent activity in the neighboring glial and neuronal compartments.

The experiment explains a mechanism of plasticity rather than claiming that every astrocyte controls every form of learning. It identifies the necessary sequence within the tested hippocampal preparation. Receptor activation, glial calcium, and postsynaptic depolarization together changed the effectiveness of later synaptic transmission.



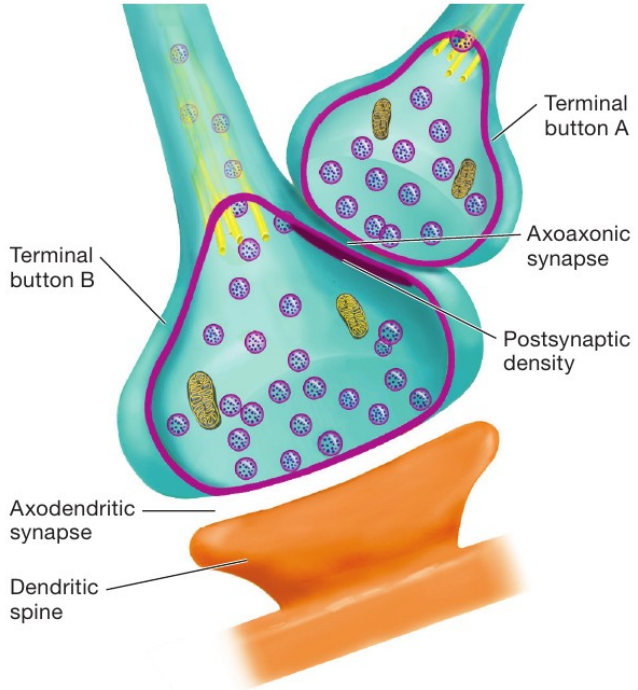
Navarrete et al. (2012), Fig. 4. Astrocytic calcium and hippocampal potentiation.

Autoreceptors regulate a neuron's own transmitter release

An **autoreceptor** responds to the transmitter released by the same neuron or neuronal population that carries it. Many autoreceptors occur on presynaptic regions. Their activation can change transmitter synthesis or release, providing feedback about the chemical signal already produced by the sending cell.

An autoreceptor is defined by which cell released the transmitter, not by being a separate universal receptor family. Different transmitters have different autoreceptors and signaling pathways. Their effects depend on the receptor, the terminal, and the cellular machinery through which receptor activation changes release.

Feedback allows released transmitter to regulate later output from the sending neuron. An **axoaxonic synapse** instead has one axon terminal acting on another terminal. Axoaxonic describes the anatomical contact, while autoreceptor describes which transmitter a receptor detects. Both can affect presynaptic communication through different arrangements.



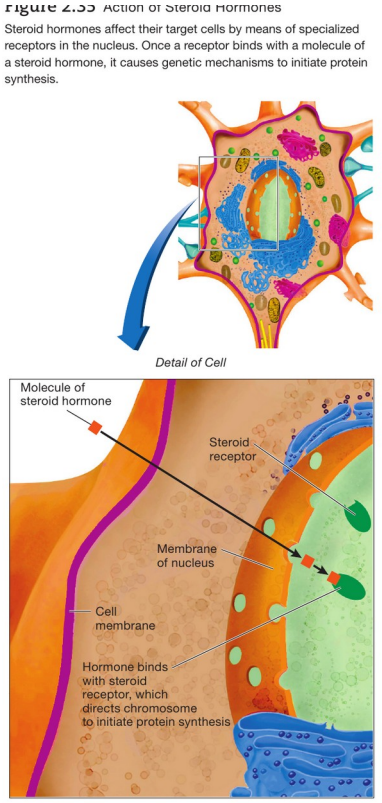
Carlson and Birkett (2022), Fig. 2.34, p. 64.

Chemical signals can act beyond one synaptic cleft

Not all chemical communication is confined to a narrow synaptic cleft. **Neuromodulators** can influence populations of neurons over broader regions, changing how those cells respond to other inputs. Their effects depend on which receptors are present and on the intracellular pathways those receptors activate.

Hormones travel through the bloodstream to reach cells carrying suitable receptors. Some hormonal signals affect neuronal function and behavior. A receptor-bearing target can respond even when it is far from the cell that secreted the hormone, distinguishing endocrine communication from release onto one nearby postsynaptic membrane.

Peptides are chains of amino acids that can also function as chemical messengers. Different messengers act at different distances and timescales. The anatomical route of delivery and the receptor mechanism together determine which cells change, how quickly the response develops, and how long the effect persists.



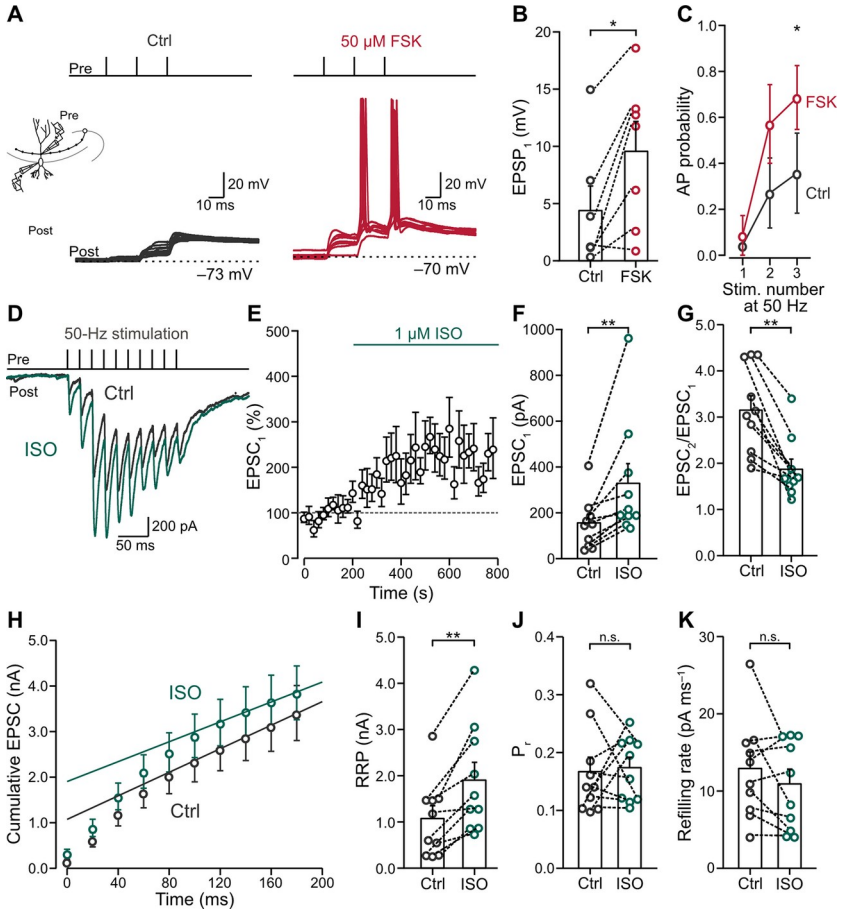
Carlson and Birkett (2022), Fig. 2.35, p. 65.

Presynaptic signaling can change release probability

The mossy-fiber study examined the **cAMP–PKA pathway**, an intracellular signaling system that can alter protein function. **cAMP** is a second messenger, and **protein kinase A** modifies proteins by phosphorylation. The investigators used stimulation and chemical manipulations to examine how this pathway strengthened presynaptic transmission.

Potentiation increased both the release-ready vesicle pool and the probability of vesicle release. The same electrical input could therefore generate a stronger chemical output. A synapse's effectiveness was adjustable because its release machinery changed state, rather than because each spike became a different type of electrical event.

The source experiments paired functional measurements with structural analysis of release sites. This combination identified specific presynaptic changes accompanying potentiation. The cellular explanation runs from intracellular signaling to altered vesicle availability and positioning, then to the postsynaptic current produced when transmitter is released.



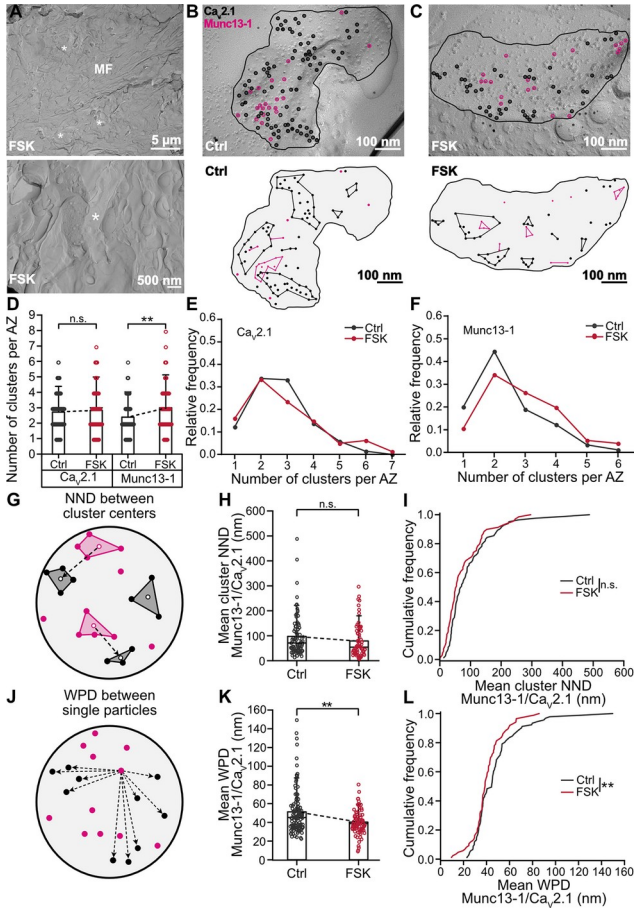
Kim et al. (2024), Fig. 2. Presynaptic vesicles and release-site organization.

Synaptic strengthening reorganizes the release site

The investigators compared molecular arrangements before and after presynaptic potentiation. Calcium-channel clusters and vesicle-priming machinery changed their spacing. More vesicles were also positioned near the membrane. Several changes therefore occurred at the small region where calcium entry triggers secretion.

The results provide a physical account of a stronger synapse. A terminal can prepare more vesicles and alter their access to the local calcium signal. It can thus change the efficiency of chemical transmission without changing which axon contacts which neuron or requiring a new anatomical connection to form.

The authors proposed that nanoscopic reorganization may contribute to information storage. That proposal distinguishes a possible memory mechanism from a direct measurement of a particular remembered event. The measured findings concern the release site's structure and function after potentiation in the hippocampal preparation.



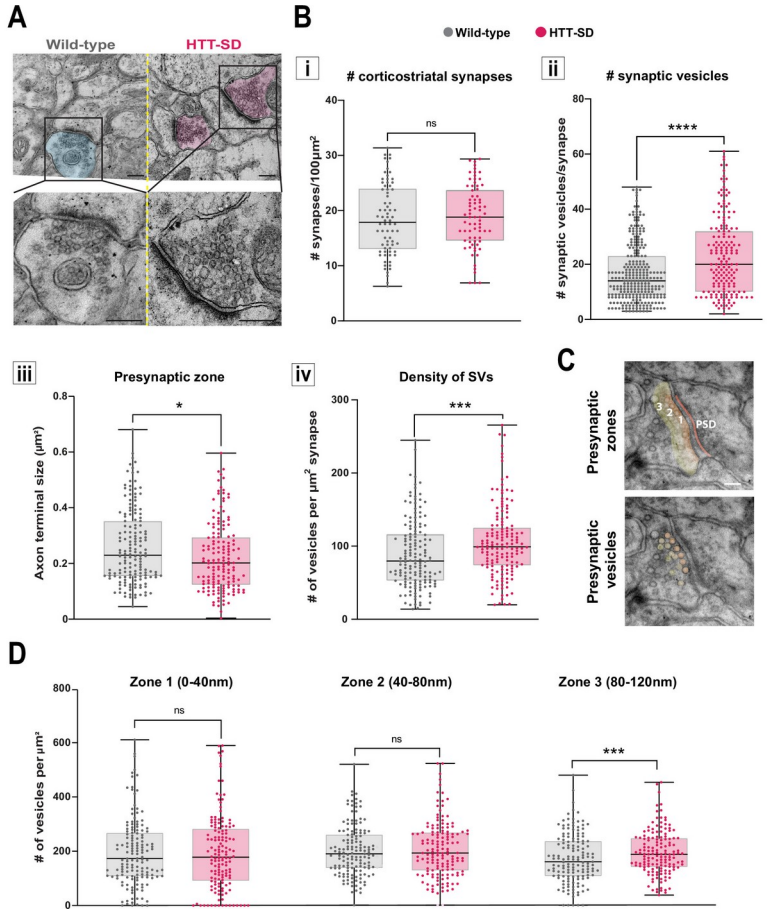
Kim et al. (2024), Fig. 4. Presynaptic vesicles and release-site organization.

Cellular transport can alter a learned motor skill

The huntingtin study followed a mechanism from axonal cargo delivery to synaptic function and behavior. Altered huntingtin phosphorylation recruited more KIF1A motor activity, increased vesicle-precursor delivery, and changed release at corticostriatal synapses. The experiments tested successive cellular events within the same investigated pathway.

Mice with the altered transport regulation showed impaired learning on a rotating-rod task. Reducing KIF1A restored both transport and motor-skill learning toward control performance. The intervention therefore tested whether excessive cargo delivery contributed to the behavioral defect rather than merely accompanying it.

The findings explain how a change inside an axon can affect the animal's performance through a specific synaptic mechanism. Movement depends on functioning circuits, and those circuits depend on membrane signaling, transmitter release, and the ongoing delivery of cellular materials that keep distant terminals supplied.



Vitet et al. (2023), Fig. 4. Axonal transport and synaptic function.

Key takeaways

Cellular compartments Dendrites receive input, axons conduct signals, and terminals release messengers. Transport maintains these distant regions.

Glial support Glia regulate the neural environment, remove debris, and construct and maintain myelin; these are distinct cellular functions.

Membrane voltage Selective permeability and ionic gradients establish voltage. Channel opening changes current; pumps maintain gradients using metabolic energy.

Action potentials Sodium-channel activation regenerates depolarization. Inactivation and potassium currents promote recovery, while myelin changes conduction.

Chemical transmission Presynaptic calcium triggers vesicle fusion. Receptor identity and ionic gradients determine the receiving cell’s response.

Adjustable communication Input timing, dendritic channels, transmitter clearance, and presynaptic organization change synaptic effectiveness and behavioral output.